

penev_B: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00001	1	1.000	$\left\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \right\rangle = 0$	$\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \rangle = 0$	from (4.57), $\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \rangle = 0$, hence $\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$ (4.44). Analogously, the
penev_B_F00002	1	1.000	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$	$\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$	from (4.57), $\langle \mathbf{s}_{\parallel}, \mathbf{s}_{\perp} \rangle = 0$, hence $\ \mathbf{s}\ ^2 = \ \mathbf{s}_{\parallel}\ ^2 + \ \mathbf{s}_{\perp}\ ^2$ (4.44). Analogously, the
penev_B_F00003	1	1.000	O_m	O_m	Notably, the change of a single O_m does not change any of the other projections
penev_B_F00004	1	0.430	$\mathbf{s}^{\text{rec}}$	$\mathbf{s}^{\text{rec}}$	of $\mathbf{s}^{\text{rec}}$ to $\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.56); this comes directly from (4.54).
penev_B_F00005	1	0.430	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	of $\mathbf{s}^{\text{rec}}$ to $\{\mathbf{q}_m\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.56); this comes directly from (4.54).
penev_B_F00006	1	0.577	$O^{\text{rec}}(\mathbf{x})$ (4.45)	$O^{\text{rec}}(\mathbf{x})$ (4.45)	Now we are prepared to calculate $O^{\text{rec}}(\mathbf{x})$ (4.45):
penev_B_F00007	1	0.606	$\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$	with the action of $\mathbf{O}^*$ on the dual-basis vectors $\{\mathbf{p}_m\}_{\mathbf{x}_m \in \mathcal{M}}$, only this time, the
penev_B_F00008	1	0.486	$\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$	$\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$	with $\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$. Therefore, $\mathbf{B} = \mathbf{Q}^{-1}$ and
penev_B_F00009	1	0.486	$\mathbf{B} = \mathbf{Q}^{-1}$	$\mathbf{B} = \mathbf{Q}^{-1}$	with $\mathbf{Q}_{nm} = \langle \mathbf{q}_n, \mathbf{q}_m \rangle$. Therefore, $\mathbf{B} = \mathbf{Q}^{-1}$ and
penev_B_F00010	1	0.686	(4.62)	(4.62)	(4.62)
penev_B_F00011	1	0.592	$\mathbf{Q}$	$\mathbf{Q}$	Notably, the matrix $\mathbf{Q}$ is the <i>restriction</i> of the matrix $\mathbf{P}$ on the set of reference
penev_B_F00012	1	0.592	$\mathbf{P}$	$\mathbf{P}$	Notably, the matrix $\mathbf{Q}$ is the <i>restriction</i> of the matrix $\mathbf{P}$ on the set of reference
penev_B_F00013	1	1.000	$\mathcal{M}$	$\mathcal{M}$	points, $\mathcal{M}$:
penev_B_F00014	1	0.826	$\mathbf{x}$	$\mathbf{x}$	Analogously to (4.48–4.49), the $\mathbf{x}$ dependence of the reconstructors $a_m(\mathbf{x})$ de-
penev_B_F00015	1	0.826	$a_m(\mathbf{x})$	$a_m(\mathbf{x})$	Analogously to (4.48–4.49), the $\mathbf{x}$ dependence of the reconstructors $a_m(\mathbf{x})$ de-
penev_B_F00016	1	1.000	ϕ	ϕ	pends only on the locations of the reference points $\mathcal{M}$, but not on ϕ itself.
penev_B_F00017	1	0.999	ϕ is	ϕ is	The reconstruction of the example ϕ is
penev_B_F00018	1	0.654	$K^{(-1)}_n(\mathbf{x}) = \left(\hat{\mathbf{x}}, K^{(-1)} \hat{\mathbf{x}}_n \right)$	$K^{(-1)}_n(\mathbf{x}) = \left(\hat{\mathbf{x}}, K^{(-1)} \hat{\mathbf{x}}_n \right)$	with $K^{(-1)}_n(\mathbf{x}) = \left(\hat{\mathbf{x}}, K^{(-1)} \hat{\mathbf{x}}_n \right)$.
penev_B_F00019	1	0.269	$ \mathcal{M} $	$ \mathcal{M} $	this might not seem natural—when $ \mathcal{M} $ is large, $\mathbf{Q}$ is also large, and inverting
penev_B_F00020	1	0.999	$\mathbf{Q}$	$\mathbf{Q}$	it is difficult. Nevertheless, $\mathbf{Q}$ is a special matrix—most of its values are van-
penev_B_F00021	1	0.623	$\{P_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{P_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$	ishingly small; because the supports of $\{P_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$ are local, $\mathbf{Q}$ is <i>sparse</i> .
penev_B_F00022	1	0.990	$\mathbf{x}_m$	$\mathbf{x}_m$	For each point $\mathbf{x}_m$, the m -th column of $\mathbf{Q}$ has substantially nonvanishing val-
penev_B_F00023	1	0.990	m	m	For each point $\mathbf{x}_m$, the m -th column of $\mathbf{Q}$ has substantially nonvanishing val-
penev_B_F00024	1	0.446	$P_m(\mathbf{x}_{mn})$	$P_m(\mathbf{x}_{mn})$	$P_m(\mathbf{x}_m)$. Moreover, this is true even when the volume of the receptor grid V
penev_B_F00025	1	0.446	V	V	$P_m(\mathbf{x}_m)$. Moreover, this is true even when the volume of the receptor grid V
penev_B_F00026	1	0.961	$\mathbf{x}_m$	$\mathbf{x}_m$	increases—adding more and more receptors distant to $\mathbf{x}_m$ does not add non-
penev_B_F00027	1	0.905	N	N	small, regardless of N and V . ³⁶
penev_B_F00028	1	0.905	V	V	small, regardless of N and V . ³⁶
penev_B_F00029	1	0.709	$\phi(\mathbf{x})$	$\phi(\mathbf{x})$	In this Section, we have shown how to represent a sensory stimulus $\phi(\mathbf{x})$
penev_B_F00030	1	1.000	$\mathcal{P}[\phi(\mathbf{x})]$	$\mathcal{P}[\phi(\mathbf{x})]$	and its probability $\mathcal{P}[\phi(\mathbf{x})]$, in the context of an ensemble with low dimension-
penev_B_F00031	1	0.968	$O^{\text{rec}}(\mathbf{x})$	$O^{\text{rec}}(\mathbf{x})$	will result in a large error in $O^{\text{rec}}(\mathbf{x})$.
penev_B_F00032	1	0.743	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	So far, we have addressed the question, given a sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$,
penev_B_F00033	1	1.000	$O(\mathbf{x})$	$O(\mathbf{x})$	how to reconstruct $O(\mathbf{x})$ and, from there, the example ϕ and its probabil-
penev_B_F00034	1	1.000	$\mathcal{P}[\phi]$	$\mathcal{P}[\phi]$	ity $\mathcal{P}[\phi]$.
penev_B_F00035	1	0.999	$\mathcal{M}^{(0)} = \emptyset$	$\mathcal{M}^{(0)} = \emptyset$	In the sparsification stage, we start with the empty set $\mathcal{M}^{(0)} = \emptyset$ and
penev_B_F00036	1	0.866	n	n	perform a series of iterative steps; at the n -th step, we add to $\mathcal{M}^{(n)}$ one reference
penev_B_F00037	1	0.866	$\mathcal{M}^{(n)}$	$\mathcal{M}^{(n)}$	perform a series of iterative steps; at the n -th step, we add to $\mathcal{M}^{(n)}$ one reference
penev_B_F00038	1	0.806	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	The resulting set $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ is an LFA representation of ϕ .
penev_B_F00039	1	1.000	$\mathcal{M} = \mathcal{M}^{(n)}$	$\mathcal{M} = \mathcal{M}^{(n)}$	First, given the current set $\mathcal{M} = \mathcal{M}^{(n)}$, we calculate the current maximum-

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penev_B_F00040	1	■ 0.479	$0^{\text{e c}}(\mathrm{x})$	$O^{\text{rec}}(\mathbf{x})$	likelihood estimate $O^{\text{rec}}(\mathbf{x})$ by (4.67–4.68) and calculate the current error
penev_B_F00041	1	■ 0.718	x_{m+1}	$\mathbf{x}_{m+1}$	Second, we look for the grid point $\mathbf{x}_{m+1}$, such that the value of $O^{\text{err}}(\mathbf{x}_{m+1})$
penev_B_F00042	1	■ 0.718	$0^{\text{err}}(\mathrm{x}_{m+1})$	$O^{\text{err}}(\mathbf{x}_{m+1})$	Second, we look for the grid point $\mathbf{x}_{m+1}$, such that the value of $O^{\text{err}}(\mathbf{x}_{m+1})$
penev_B_F00043	1	■ 0.980	$0^{\text{err}}(\mathrm{bf{x}_{m+1}})^2$	$O^{\text{err}}(\mathbf{x}_{m+1})^2$	is maximum. If its square, $O^{\text{err}}(\mathbf{x}_{m+1})^2$, is below a predetermined threshold,
penev_B_F00044	1	■ 1.000	$-\log \mathcal{P}[\phi]$	$-\log \mathcal{P}[\phi]$	terion. We will remember that the information content of the example, $-\log \mathcal{P}[\phi]$, is de-
penev_B_F00045	1	■ 0.810	$-\log \mathcal{P}[\phi] - (-\log \mathcal{P}[\phi^{\text{rec}}])$	$-\log \mathcal{P}[\phi] - (-\log \mathcal{P}[\phi^{\text{rec}}])$	$-\log \mathcal{P}[\phi] - (-\log \mathcal{P}[\phi^{\text{rec}}])$, is equal to $\ O^{\text{err}}\ _F^2$. Therefore, by bounding of the maximum
penev_B_F00046	1	■ 0.810	$0^{\text{err}}(\mathrm{x}_{m+1})$	$\ O^{\text{err}}\ _U^2$	$-\log \mathcal{P}[\phi] - (-\log \mathcal{P}[\phi^{\text{rec}}])$, is equal to $\ O^{\text{err}}\ _F^2$. Therefore, by bounding of the maximum
penev_B_F00047	1	■ 0.771	$0^{\text{err}}(\mathrm{x}_{m+1})$	$\ O^{\text{err}}\ _E^2 = \frac{1}{V} \sum_x O^{\text{err}}(\mathbf{x})^2 (2.8)$	term in the sum $\ O^{\text{err}}\ _E^2 = \frac{1}{V} \sum_x O^{\text{err}}(\mathbf{x})^2 (2.8)$, we bound the whole sum and, therefore,
penev_B_F00048	1	■ 0.779	$\mathrm{bf{x}_{m+1}}$	$\mathbf{x}_{m+1}$	Third, if the algorithm has not terminated yet, we add $\mathbf{x}_{m+1}$ to $\mathcal{M}^{(n)}$, which
penev_B_F00049	1	■ 0.779	$\mathcal{M}^{(n)}$	$\mathcal{M}^{(n)}$	Third, if the algorithm has not terminated yet, we add $\mathbf{x}_{m+1}$ to $\mathcal{M}^{(n)}$, which
penev_B_F00050	1	■ 0.974	$\mathcal{M}^{(n+1)}$	$\mathcal{M}^{(n+1)}$	results in $\mathcal{M}^{(n+1)}$, and go to step $n + 1$. ³⁹
penev_B_F00051	1	■ 0.974	$n + 1$	$n + 1$	results in $\mathcal{M}^{(n+1)}$, and go to step $n + 1$. ³⁹
penev_B_F00052	1	■ 0.978	$ \mathcal{M} = N$	$ \mathcal{M} = N$	dicted well, therefore they are linearly independent, and by the time $ \mathcal{M} = N$,
penev_B_F00053	1	■ 0.972	$0^{\text{err}}(x) = 0$	$O^{\text{err}}(x) = 0$	$O^{\text{err}}(\mathbf{x}) = 0$ (4.36).
penev_B_F00054	1	■ 1.000	$\mathcal{M}^{(25)}$	$\mathcal{M}^{(25)}$	The location and the order of first 25 points, those in $\mathcal{M}^{(25)}$ (Fig. 4.4a),
penev_B_F00055	1	■ 0.990	$\mathcal{M}^{(64)}$	$\mathcal{M}^{(64)}$	which is a desirable property. The points in $\mathcal{M}^{(64)}$ enough for acceptable
penev_B_F00056	1	■ 0.822	a_r	a_r	number of values 64. We calculated the PCA coefficients a_r (2.47–2.52) and
penev_B_F00057	1	■ 0.996	$\mathbf{x}_{m+1}$	$\mathbf{x}_{m+1}$	footnote 39 we argue that this is a costly calculation. Moreover, if we add the point $\mathbf{x}_{m+1}$,
penev_B_F00058	1	■ 0.958	$0^{\text{err}}(\mathrm{bf{x}_{m+1}})$	$O^{\text{err}}(\mathbf{x}_{m+1})$	we are making a correction, $O^{\text{err}}(\mathbf{x}_{m+1})$, to the estimated value $O^{\text{rec}}(\mathbf{x}_{m+1})$; this has to be
penev_B_F00059	1	■ 0.958	$0^{\text{rec}}(\mathrm{bf{x}_{m+1}})$	$O^{\text{rec}}(\mathbf{x}_{m+1})$	we are making a correction, $O^{\text{err}}(\mathbf{x}_{m+1})$, to the estimated value $O^{\text{rec}}(\mathbf{x}_{m+1})$; this has to be
penev_B_F00060	1	■ 0.637	$O(\mathrm{bf{x}_{m+1}})$	$O(\mathbf{x}_{m+1})$	compared to the error in the measurement of $O(\mathbf{x}_{m+1})$ itself, so a criterion defined at the
penev_B_F00061	1	■ 0.938	$0^{\text{err}}(\mathrm{x}_{m+1})$	$O^{\text{err}}(\mathbf{x}_{m+1})$	level of the individual error, $O^{\text{err}}(\mathbf{x}_{m+1})$, might not be such a bad idea after all. Also, we
penev_B_F00062	1	■ 0.515	$0^{\text{err}}(\mathrm{x}_{m+1})$	$\ O^{\text{err}}\ _U^2$	provisionally taken; $\mathbf{Q}$, inverted; the resulting $\ O^{\text{err}}\ _U^2$, stored; and, after all $\sim V$ provisional
penev_B_F00063	1	■ 0.515	$\sim V$	$\sim V$	provisionally taken; $\mathbf{Q}$, inverted; the resulting $\ O^{\text{err}}\ _U^2$, stored; and, after all $\sim V$ provisional
penev_B_F00064	2	■ 0.869	$\mathbb{I}$	$\mathbb{I}$	The building of the mask $\mathcal{M}$ for Example 1, reconstructed in the context of
penev_B_F00065	2	■ 1.000	$N=220, n=\sigma_{400}$	$N = 220, n = \sigma_{400}$	Ensemble 1. $N = 220, n = \sigma_{400}$ (see Fig. 4.1). a : the first 25 points— $\mathcal{M}^{(25)}$,
penev_B_F00066	2	■ 1.000	a	a	Ensemble 1. $N = 220, n = \sigma_{400}$ (see Fig. 4.1). a : the first 25 points— $\mathcal{M}^{(25)}$,
penev_B_F00067	2	■ 0.999	$\phi(\mathrm{bf{x})}$	$\phi(\mathbf{x})$	overlayed on $\phi(\mathbf{x})$ and numbered sequentially. (The label for point 11, in the
penev_B_F00068	2	■ 0.999	b	b	is the bottommost point on the right edge, shown on b). b : the points in $\mathcal{M}^{(64)}$.
penev_B_F00069	2	■ 0.995	$N=220$	$N = 220$	the LFA ones $O(\mathbf{x})$ (4.31) for $N = 220$, then sparsified to get $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$.
penev_B_F00070	2	■ 0.995	$\phi^{\text{rec}}(\mathrm{bf{x})}$	$\phi^{\text{rec}}(\mathbf{x})$	We calculated the reconstruction $\phi^{\text{rec}}(\mathbf{x})$ either (a), from (2.50) with $N = 64$, or
penev_B_F00071	2	■ 0.995	$N=64$	$N = 64$	We calculated the reconstruction $\phi^{\text{rec}}(\mathbf{x})$ either (a), from (2.50) with $N = 64$, or
penev_B_F00072	2	■ 0.965	8×8	8×8	For comparison, we subsampled $\phi(\mathbf{x})$ on a uniform 8×8 grid and reconstructed
penev_B_F00073	2	■ 1.000	Ψ_1	Ψ_1	it with those 64 points (c). In all cases we reconstructed about Ψ_1 the average
penev_B_F00074	2	■ 1.000	a_1	a_1	of the ensemble; we kept a_1 fixed. Evidentially, the perceptual quality of the
penev_B_F00075	2	■ 0.999	$\phi^{\text{rec}}(\mathrm{bf{x})}$	$\phi^{\text{rec}}(\mathbf{x})$	reconstruction $\phi^{\text{rec}}(\mathbf{x})$ (top row) is the best, and the error $\phi(\mathbf{x}) - \phi^{\text{rec}}(\mathbf{x})$ (bot-
penev_B_F00076	2	■ 0.999	$\phi(\mathrm{bf{x})} - \phi^{\text{rec}}(\mathrm{bf{x})}$	$\phi(\mathbf{x}) - \phi^{\text{rec}}(\mathbf{x})$	reconstruction $\phi^{\text{rec}}(\mathbf{x})$ (top row) is the best, and the error $\phi(\mathbf{x}) - \phi^{\text{rec}}(\mathbf{x})$ (bot-
penev_B_F00077	2	■ 0.499	c	c	and 508 for (a), (b), and (c), respectively, out of 560 total power in $(\phi(\mathbf{x}) - a_1 \Psi_1)$. Notably,
penev_B_F00078	2	■ 0.499	$(\phi(\mathrm{x}) - a_1 \mathrm{bf{x}}) \Psi_1$	$(\phi(\mathbf{x}) - a_1 \Psi_1)$	and 508 for (a), (b), and (c), respectively, out of 560 total power in $(\phi(\mathbf{x}) - a_1 \Psi_1)$. Notably,
penev_B_F00079	2	■ 0.990	Ψ_2	Ψ_2	as Ψ_2 on Fig. 3.9. Indeed, the m.s.e. due to the error in a_2 in (b) is 75. If one fixes not
penev_B_F00080	2	■ 0.990	a_2	a_2	as Ψ_2 on Fig. 3.9. Indeed, the m.s.e. due to the error in a_2 in (b) is 75. If one fixes not
penev_B_F00081	2	■ 0.474	$0^{\text{err}}(\mathrm{bf{x}_{m+1}})$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.674.68)	(2.48). b : reconstruction from the sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.67–4.68) with
penev_B_F00082	2	■ 1.000	$\mathcal{M} = \mathcal{M}^{(64)}$	$\mathcal{M} = \mathcal{M}^{(64)}$	$\mathcal{M} = \mathcal{M}^{(64)}$ shown on Fig. 4.4b, which was produced by the serial algorithm
penev_B_F00083	2	■ 0.788	ϕ_{64}^{err}	ϕ_{64}^{err}	The error on (a) is exactly equal to ϕ_{64}^{err} , and the error on (b) is perceptually
penev_B_F00084	2	■ 0.830	ϕ_{100}^{err}	ϕ_{100}^{err}	somewhere between ϕ_{100}^{err} and ϕ_{200}^{err} (cf. also the reconstruction with ϕ_{200}^{err}), both
penev_B_F00085	2	■ 0.830	ϕ_{200}^{err}	ϕ_{200}^{err}	somewhere between ϕ_{100}^{err} and ϕ_{200}^{err} (cf. also the reconstruction with ϕ_{200}^{err}), both
penev_B_F00086	2	■ 0.830	ϕ_{200}^{rec}	ϕ_{200}^{rec}	somewhere between ϕ_{100}^{err} and ϕ_{200}^{err} (cf. also the reconstruction with ϕ_{200}^{rec}), both
penev_B_F00087	2	■ 0.992	$5 \times$	$5 \times$	are magnified $5 \times$ (cf. Fig. 3.12, Fig. 4.3). The error on (c) is not magnified.
penev_B_F00088	2	■ 1.000	$(m-1)$	$(m-1)$	the $(m-1)$ -st step through a simple algebraic formula. The algorithm for inversion through
penev_B_F00089	2	■ 0.585	$ \mathcal{M} 20\% - 25\%$	$ \mathcal{M} 20\% - 25\%$	ence points $ \mathcal{M} $ 20%–25% of their respective PCA dimensionalities N ; 99%,
penev_B_F00090	2	■ 0.585	$N ; 99\%$	$N ; 99\%$	ence points $ \mathcal{M} $ 20%–25% of their respective PCA dimensionalities N ; 99%,
penev_B_F00091	2	■ 0.970	$45\% - 50\%$	$45\% - 50\%$	with 45%–50%. Therefore, the necessary LFA dimensionality for a given “in-
penev_B_F00092	2	■ 0.560	99%	99%	to get 99% of the fidelity?—the incremental benefit of adding a reference point in the 50%
penev_B_F00093	2	■ 0.560	50%	50%	to get 99% of the fidelity?—the incremental benefit of adding a reference point in the 50%

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penev_B_F0094	2	■ 1.000	r	r	LFA representations by taking the sums in (4.29) in several separate bands for r , instead of
penev_B_F0095	2	■ 1.000	≈ 1	≈ 1	contributes on the average ≈ 1 .
penev_B_F0096	2	■ 0.999	$N \in$	$N \in$	the number of referece points $ \mathcal{M} $ is shown with dashed lines for $N \in$
penev_B_F0097	3	■ 0.977	$S - N$	$S - N$	Figure 4.7: Scaling of LFA S - N Diagrams
penev_B_F0098	3	■ 0.682	$\ O\ _U^2 = \ \Phi^{*+}\phi\ _S^2$	$\ O\ _U^2 = \ \Phi^{*+}\phi\ _S^2$	$\ O\ _U^2 = \ \Phi^{*+}\phi\ _S^2$ (4.22), as a function of $ \mathcal{M} /N$, the relative dimensionalities
penev_B_F0099	3	■ 0.682	$ \mathcal{M} / N$	$ \mathcal{M} /N$	$\ O\ _U^2 = \ \Phi^{*+}\phi\ _S^2$ (4.22), as a function of $ \mathcal{M} /N$, the relative dimensionalities
penev_B_F00100	3	■ 0.977	$N \in \{600, 400, 300, 220, 110, 64, 32\}$	$N \in \{600, 400, 300, 220, 110, 64, 32\}$	of the LFA and PCA representations for $N \in \{600, 400, 300, 220, 110, 64, 32\}$
penev_B_F00101	3	■ 1.000	$\mathcal{M}^{(50)}$	$\mathcal{M}^{(50)}$	on Fig. 4.8. Almost all of the points in the respective sets $\mathcal{M}^{(50)}$ are on the
penev_B_F00102	3	■ 0.496	$\{O(x_m)\}_{x_m \in \mathcal{M}}$	$\{O(x_m)\}_{x_m \in \mathcal{M}}$	given a sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ of the dense LFA outputs $O(\mathbf{x})$ for an
penev_B_F00103	3	■ 0.496	$O(\mathbf{x})$	$O(\mathbf{x})$	given a sparsification $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ of the dense LFA outputs $O(\mathbf{x})$ for an
penev_B_F00104	3	■ 0.951	$\{a_m(\mathbf{x})\}_{x_m \in \mathcal{M}}$	$\{a_m(\mathbf{x})\}_{x_m \in \mathcal{M}}$	given by the set of interpolating reconstructors $\{a_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.68).
penev_B_F00105	3	■ 0.342	$\sum_{n=1}^{\mathcal{M}} Q^{-1}_{nnn} K^{(-1)}_n(\mathbf{x})$	$\sum_{n=1}^{\mathcal{M}} Q^{-1}_{nnn} K^{(-1)}_n(\mathbf{x})$	combination, only with a different set of reconstructors, $\sum_{n=1}^{\mathcal{M}} Q^{-1}_{nnn} K^{(-1)}_n(\mathbf{x})$.
penev_B_F00106	3	■ 1.000	$V=8640$	$V = 8640$	in the context of Ensemble 3 (see Appendix A.1 for definition) with $V = 8640$,
penev_B_F00107	3	■ 1.000	$T=1038, N=400, n=\sigma_{400}$	$T = 1038, N = 400, n = \sigma_{400}$	$T=1038, N=400, n=\sigma_{400}$ (see Fig. 4.1).
penev_B_F00108	3	■ 0.938	$\mathbf{Q}^{-1}$	$\mathbf{Q}^{-1}$	operation. By the same argument, $\mathbf{Q}^{-1}$ will be sparse, and acting with it on O_m
penev_B_F00109	3	■ 0.907	$\mathcal{O}_I$	$\mathcal{O}_I$	will be a local operation too. Moreover, one may try to solve for $\mathcal{O}_I$ directly
penev_B_F00110	3	■ 1.000	$\mathbf{Q}^{-1}$	$\mathbf{Q}^{-1}$	and never explicitly store $\mathbf{Q}^{-1}$, which will eliminate one computational step
penev_B_F00111	3	■ 0.644	$\mathcal{O}_1$	$\mathcal{O}_1$	Some intuition into what is involved in the calculation of $\mathcal{O}_1$ may be de-
penev_B_F00112	3	■ 1.000	$\mathcal{O}_l$	$\mathcal{O}_l$	may be solved for $\mathcal{O}_l$ (cf. 4.72) by a rapid iteration.
penev_B_F00113	4	■ 0.601	$G(\mathcal{O})(\mathbf{x}) = g(\mathcal{O}(\mathbf{x}))$	$G(\mathcal{O})(\mathbf{x}) = g(\mathcal{O}(\mathbf{x}))$	where $G(\mathcal{O})(\mathbf{x}) = g(\mathcal{O}(\mathbf{x}))$ with g , a suitably chosen <i>sigmoid</i> function, ⁴⁶ and
penev_B_F00114	4	■ 1.000	τ	τ	τ is the characteristic time scale.
penev_B_F00115	4	■ 1.000	$\mathcal{O}(\mathbf{x})$	$\mathcal{O}(\mathbf{x})$	to pass the large ones intact. Then, if all $\mathcal{O}(\mathbf{x})$ either vanish, or are sufficiently
penev_B_F00116	4	■ 0.555	$\frac{d}{dt} \mathcal{O}(\mathbf{x}) = 0$	$\frac{d}{dt} \mathcal{O}(\mathbf{x}) = 0$	large, g will be the identity. In this regime, for the steady states, $\frac{d}{dt} \mathcal{O}(\mathbf{x}) = 0$,
penev_B_F00117	4	■ 0.465	$O^{\text{rec}} = \mathbf{P}\mathcal{O} = \mathbf{K}\phi = O$	$O^{\text{rec}} = \mathbf{P}\mathcal{O} = \mathbf{K}\phi = O$	The steady-state equation (4.76) is equivalent to $O^{\text{rec}} = \mathbf{P}\mathcal{O} = \mathbf{K}\phi = O$
penev_B_F00118	4	■ 0.997	$\{\mathcal{O}_m\}_{x_m \in \mathcal{M}}$	$\{\mathcal{O}_m\}_{x_m \in \mathcal{M}}$	hence, the sparsified outputs $\{\mathcal{O}_m\}_{\mathbf{x}_m \in \mathcal{M}}$ are solutions of (4.76) and, therefore,
penev_B_F00119	4	■ 0.840	$g: \mathcal{R} \rightarrow [a, b]$	$g: \mathcal{R} \rightarrow [a, b]$	⁴⁶ A sigmoid function is a monotonic nondecreasing function $g: \mathcal{R} \rightarrow [a, b]$, often $a = -b$,
penev_B_F00120	4	■ 0.840	$a = -b$	$a = -b$	⁴⁶ A sigmoid function is a monotonic nondecreasing function $g: \mathcal{R} \rightarrow [a, b]$, often $a = -b$,
penev_B_F00121	4	■ 0.809	g'	g'	and usually g' is non-zero only near the origin, or a suitably chosen <i>threshold</i> θ .
penev_B_F00122	4	■ 0.809	θ	θ	and usually g' is non-zero only near the origin, or a suitably chosen <i>threshold</i> θ .
penev_B_F00123	4	■ 1.000	$\{\mathcal{O}(\mathbf{u})\}_{\mathbf{u} \in V}$	$\{\mathcal{O}(\mathbf{u})\}_{\mathbf{u} \in V}$	The input layer stores the example $\phi(\mathbf{x})$; the output layer, $\{\mathcal{O}(\mathbf{u})\}_{\mathbf{u} \in V}$.
penev_B_F00124	4	■ 0.803	$\mathbf{u}$	$\mathbf{u}$	The output layer unit centered at $\mathbf{u}$ has a feed-forward receptive field $\mathbf{K}_\mathbf{u}(\mathbf{x})$
penev_B_F00125	4	■ 0.803	$\mathbf{K}_\mathbf{u}(\mathbf{x})$	$\mathbf{K}_\mathbf{u}(\mathbf{x})$	The output layer unit centered at $\mathbf{u}$ has a feed-forward receptive field $\mathbf{K}_\mathbf{u}(\mathbf{x})$
penev_B_F00126	4	■ 0.998	$O^{\text{err}}(\mathbf{u})$	$O^{\text{err}}(\mathbf{u})$	The current error signal $O^{\text{err}}(\mathbf{u})$ (4.71) is calculated from the current output
penev_B_F00127	4	■ 0.962	$\mathcal{O}(\mathbf{u})$	$\mathcal{O}(\mathbf{u})$	variables $\mathcal{O}(\mathbf{u})$ by (4.72); the residual correlations $\mathbf{P}_\mathbf{u}(\mathbf{v})$ (cf. Fig. 4.1 4.2) serve
penev_B_F00128	4	■ 0.962	$\mathbf{P}_\mathbf{u}(\mathbf{v})$	$\mathbf{P}_\mathbf{u}(\mathbf{v})$	variables $\mathcal{O}(\mathbf{u})$ by (4.72); the residual correlations $\mathbf{P}_\mathbf{u}(\mathbf{v})$ (cf. Fig. 4.1 4.2) serve
penev_B_F00129	4	■ 0.772	$\mathcal{O}(\mathbf{u}) = 0$	$\mathcal{O}(\mathbf{u}) = 0$	to $\mathcal{O}(\mathbf{u}) = 0$ everywhere on the grid, with the exception of $\{\mathcal{O}_m\}_{m=1}^{ \mathcal{M} }$ (4.74).
penev_B_F00130	4	■ 0.772	$\{\mathcal{O}_m\}_{m=1}^{ \mathcal{M} }$	$\{\mathcal{O}_m\}_{m=1}^{ \mathcal{M} }$	to $\mathcal{O}(\mathbf{u}) = 0$ everywhere on the grid, with the exception of $\{\mathcal{O}_m\}_{m=1}^{ \mathcal{M} }$ (4.74).
penev_B_F00131	4	■ 0.800	g	g	If output dynamic range equalization is required, g may be chosen to preclude
penev_B_F00132	4	■ 0.607	$\mathcal{O}_i$	$\mathcal{O}_i$	large $\mathcal{O}_i$.
penev_B_F00133	4	■ 0.863	$\mathcal{P}_{\text{objects}}$	$\mathcal{P}_{\text{objects}}$	ability distribution, $\mathcal{P}_{\text{objects}}$, is entirely contained in the probability distribution
penev_B_F00134	4	■ 0.992	$\mathcal{P}_{\text{images}}$	$\mathcal{P}_{\text{images}}$	of natural images, $\mathcal{P}_{\text{images}}$.
penev_B_F00135	4	■ 0.445	$\mathcal{P}_{\text{object can}}$	$\mathcal{P}_{\text{object can}}$	approximation and parameterization. In that sense, $\mathcal{P}_{\text{objects}}$ can be considered
penev_B_F00136	5	■ 0.945	$\{I^t\}_{t \in T}$	$\{I^t\}_{t \in T}$	of images $\{I^t\}_{t \in T}$ was prepared, wherein each image contained a human face
penev_B_F00137	5	■ 1.000	I^t	I^t	in a relatively frontal pose. For each image in the ensemble, I^t , the locations
penev_B_F00138	5	■ 1.000	(X_l^t, Y_l^t)	(X_l^t, Y_l^t)	(X_l^t, Y_l^t) and (X_r^t, Y_r^t) of the left and right eye, respectively (see discussion in
penev_B_F00139	5	■ 1.000	(X_r^t, Y_r^t)	(X_r^t, Y_r^t)	(X_l^t, Y_l^t) and (X_r^t, Y_r^t) of the left and right eye, respectively (see discussion in
penev_B_F00140	5	■ 0.698	θ^t	θ^t	Subsequently, the image I^t was <i>rotated</i> , by the angle θ^t , to line the eyes
penev_B_F00141	5	■ 1.000	S^t	S^t	horizontally; <i>scaled</i> , by the scale factor S^t , to bring the inter-eye distance to a
penev_B_F00142	5	■ 0.998	X^t	X^t	pre-defined length; and <i>translated</i> , horizontally by X^t and vertically by Y^t , to
penev_B_F00143	5	■ 0.998	Y^t	Y^t	pre-defined length; and <i>translated</i> , horizontally by X^t and vertically by Y^t , to
penev_B_F00144	5	■ 0.998	ϕ^t	ϕ^t	for a given cropping, the example ϕ^t depends on the image I^t , as well as on the
penev_B_F00145	5	■ 0.827	H	H	where H is the (highly nonlinear) operator of symmetry-breaking.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00146	5	■ 0.554	$\left(X_{\{l\}}, Y_{\{l\}}\right)$	(X_l, Y_l)	eye locations, (X_l, Y_l) and $(\overline{X_r}, \overline{Y_r})$, lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00147	5	■ 0.554	$\left(X_{\{r\}}, Y_{\{r\}}\right)$	(X_r, Y_r)	eye locations, (X_l, Y_l) and (X_r, Y_r) , lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00148	5	■ 0.554	α	α	eye locations, (X_l, Y_l) and (X_r, Y_r) , lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00149	5	■ 0.554	$H\left(I^{\{t\}} ; \alpha^{\{t\}}\right)$	$H\left(I^t; \alpha^t\right)$	eye locations, (X_l, Y_l) and (X_r, Y_r) , lead to errors in α . Also, if $H(I^t; \alpha^t)$ is in
penev_B_F00150	5	■ 1.000	$H\left(I^{\{t\}} ; \alpha^{\{t\}}+\Delta \alpha\right)$	$H\left(I^t; \alpha^t+\Delta \alpha\right)$	the ensemble of faces, then $H\left(I^t; \alpha^t+\Delta \alpha\right)$ will also be in it, provided that $\Delta \alpha$
penev_B_F00151	5	■ 1.000	$\Delta \alpha$	$\Delta \alpha$	the ensemble of faces, then $H\left(I^t; \alpha^t+\Delta \alpha\right)$ will also be in it, provided that $\Delta \alpha$
penev_B_F00152	5	■ 0.951	$6(T=10944, V=3840)$	$6(T=10944, V=3840)$	Figure 5.1: PCA Analysis of Ensemble 6 ($T=10944, V=3840$)
penev_B_F00153	5	■ 1.000	$\left\{\Psi_r(\mathbf{x})\right\}_{r=1}^{16}$	$\left\{\Psi_r(\mathbf{x})\right\}_{r=1}^{16}$	The first 16 eigenmodes $\left\{\Psi_r(\mathbf{x})\right\}_{r=1}^{16}$ of Ensemble 6 (top and middle) and eight
penev_B_F00154	5	■ 0.998	2 A	2A	This figure is reproduced on Plate 2A. For comparison, part of Fig. 3.9 is
penev_B_F00155	5	■ 1.000	2 B	2B	reproduced on Plate 2B.
penev_B_F00156	5	■ 1.000	Ψ_8	Ψ_8	Notably, the first eyebrow mode is now very strong Ψ_8 . This is due to
penev_B_F00157	5	■ 1.000	Ψ_{12}	Ψ_{12}	A very interesting effect on the asymmetric Ψ_{12} (Fig. 3.9), the first strong
penev_B_F00158	5	■ 1.000	Ψ_{13}	Ψ_{13}	feature therein, can be observed—it has moved one mode down to Ψ_{13} in the
penev_B_F00159	5	■ 1.000	Ψ_{14}	Ψ_{14}	symmetrized ensemble, has weakened, and has spread to two more modes: Ψ_{14}
penev_B_F00160	5	■ 1.000	Ψ_{18}	Ψ_{18}	and Ψ_{18} . The same has happened to Ψ_{14} , the “Nixon” mode (footnote 11 on
penev_B_F00161	5	■ 1.000	$\Psi_{14}, \Psi_{15}, \Psi_{16}$	$\Psi_{14}, \Psi_{15}, \Psi_{16}$	page 34) weak vestiges of it can be observed on $\Psi_{14}, \Psi_{15}, \Psi_{16}$, and Ψ_{18} .
penev_B_F00162	5	■ 0.439	$\boldsymbol{\Psi}_1$	$\boldsymbol{\Psi}_1$	they are thin—it is evident from Ψ_1 and Ψ_{13} that the distance from the eyebrows to the eyes
penev_B_F00163	5	■ 0.439	$\boldsymbol{\Psi}_{13}$	$\boldsymbol{\Psi}_{13}$	they are thin—it is evident from Ψ_1 and Ψ_{13} that the distance from the eyebrows to the eyes
penev_B_F00164	5	■ 0.833	$\boldsymbol{\Psi}_8$	$\boldsymbol{\Psi}_8$	eyebrow location. This degree of freedom is what is observed on Ψ_8 .
penev_B_F00165	5	■ 1.000	Ψ_{100}	Ψ_{100}	The feature localization problem is very clearly evident in Ψ_{100} and Ψ_{200} .
penev_B_F00166	5	■ 1.000	Ψ_{200}	Ψ_{200}	The feature localization problem is very clearly evident in Ψ_{100} and Ψ_{200} .
penev_B_F00167	5	■ 0.935	Ψ_{500}	Ψ_{500}	Interestingly, Ψ_{500} and Ψ_{1000} show quite discernible face content; even Ψ_{2000} ,
penev_B_F00168	5	■ 0.935	Ψ_{1000}	Ψ_{1000}	Interestingly, Ψ_{500} and Ψ_{1000} show quite discernible face content; even Ψ_{2000} ,
penev_B_F00169	5	■ 0.935	Ψ_{2000}	Ψ_{2000}	Interestingly, Ψ_{500} and Ψ_{1000} show quite discernible face content; even Ψ_{2000} ,
penev_B_F00170	5	■ 1.000	$N>60$	$N>60$	is better for any $N>60$. Also, the respective perceptual identity information
penev_B_F00171	5	■ 0.955	ϕ_{1000}^{err}	ϕ_{1000}^{err}	in the error is smaller, almost nonexistent in ϕ_{1000}^{err} .
penev_B_F00172	5	■ 0.987	$S-U$	$S-U$	$S-U$ diagrams on Fig. 5.3. Not only does the symmetrized ensemble have
penev_B_F00173	5	■ 0.994	$N \approx 300$	$N \approx 300$	Whereas the entropic cost is rapidly growing after $N \approx 300$ for the asym-
penev_B_F00174	5	■ 1.000	$N \in [200, 500]$	$N \in [200, 500]$	metric ensemble, it is relatively flat in the regime $N \in [200, 500]$ for the sym-
penev_B_F00175	6	■ 1.000	ϕ_N^{rec}	ϕ_N^{rec}	The successive reconstructions ϕ_N^{rec} labeled with their SNR (top) and their
penev_B_F00176	6	■ 0.999	ϕ_N^{err}	ϕ_N^{err}	respective residual errors ϕ_N^{err} (bottom) for Example 1 (Fig. 3.10- <i>original</i>).
penev_B_F00177	6	■ 0.923	$\{ }^6 \text{cf. } \Psi_8$	$\{ }^6 \text{cf. } \Psi_8$	$\{ }^6 \text{cf. } \Psi_8$ on Fig. 5.1
penev_B_F00178	6	■ 0.983	$T=10944$	$T=10944$	Figure 5.3: $S-U$ diagram of a reconstruction with Ensemble 6 ($T=10944$,
penev_B_F00179	6	■ 0.965	$V=3840$	$V=3840$	$V=3840$)
penev_B_F00180	6	■ 0.983	$\text{SNR} \approx \infty$	$\text{SNR} \approx \infty$	The $S-U$ diagram of Ensemble 6 saturates to $\text{SNR} \approx \infty$ very quickly after
penev_B_F00181	6	■ 0.793	≈ 16	≈ 16	$\text{SNR} \approx 16$, leveling off at $\langle S \rangle = 2.35$, thereby forming a relatively smooth
penev_B_F00182	6	■ 0.793	$\langle S \rangle = 2.35$	$\langle S \rangle = 2.35$	$\text{SNR} \approx 16$, leveling off at $\langle S \rangle = 2.35$, thereby forming a relatively smooth
penev_B_F00183	6	■ 0.849	≈ 7	≈ 7	S-shaped curve after $\text{SNR} \approx 7$.
penev_B_F00184	6	■ 0.991	$\mathbf{x}_0$	$\mathbf{x}_0$	is marked with the respective positions $\mathbf{x}_0$ for $a \in \mathcal{A}$.
penev_B_F00185	6	■ 0.991	$a \in \mathcal{A}$	$a \in \mathcal{A}$	is marked with the respective positions $\mathbf{x}_0$ for $a \in \mathcal{A}$.
penev_B_F00186	6	■ 0.941	e	e	$a \in \mathcal{A}$: Five receptive fields $K(\mathbf{x}_0, \mathbf{y}) = \mathbf{K}_{\mathbf{x}_0}(\mathbf{y}) = (\mathbf{K}, \hat{\mathbf{x}}_0)$ (cf. 4.25) (top row)
penev_B_F00187	6	■ 0.941	$K(\mathbf{x}_0, \mathbf{y}) = \mathbf{K}_{\mathbf{x}_0}(\mathbf{y}) = (\mathbf{K}, \hat{\mathbf{x}}_0)$	$K(\mathbf{x}_0, \mathbf{y}) = \mathbf{K}_{\mathbf{x}_0}(\mathbf{y}) = (\mathbf{K}, \hat{\mathbf{x}}_0)$	$a \in \mathcal{A}$: Five receptive fields $K(\mathbf{x}_0, \mathbf{y}) = \mathbf{K}_{\mathbf{x}_0}(\mathbf{y}) = (\mathbf{K}, \hat{\mathbf{x}}_0)$ (cf. 4.25) (top row)
penev_B_F00188	6	■ 1.000	$\mathbf{P}_{\mathbf{x}_0}(\mathbf{y})$	$\mathbf{P}_{\mathbf{x}_0}(\mathbf{y})$	and correlators $\mathbf{P}_{\mathbf{x}_0}(\mathbf{y})$ (middle row) at the five choices for $\mathbf{x}_0$. The parameters
penev_B_F00189	6	■ 0.999	$(T=10944, V=3840), N=400$	$(T=10944, V=3840), N=400$	are $(T=10944, V=3840), N=400$, and $n=\sigma_{400}$ (see Fig. 4.1).
penev_B_F00190	6	■ 0.999	$n=\sigma_{400}$	$n=\sigma_{400}$	are $(T=10944, V=3840), N=400$, and $n=\sigma_{400}$ (see Fig. 4.1).
penev_B_F00191	6	■ 0.988	$\{ \mathbf{x}_0 \}$	$\{ \mathbf{x}_0 \}$	The locations $\{ \mathbf{x}_0 \}$ were chosen to be the same as those on Fig. 4.1, whose
penev_B_F00192	6	■ 0.974	$H(I; \alpha)$ (5.2)	$H(I; \alpha)$ (5.2)	$H(I; \alpha)$ (5.2) on the set of images $\{ I^t \}_{t \in T}$. In the restricted sense, used for the
penev_B_F00193	6	■ 0.974	$\{ \alpha^t \}_{t \in T}$ (5.1), $\{ }^8$	$\{ \alpha^t \}_{t \in T}$ (5.1), $\{ }^8$	of coefficients $\{ \alpha^t \}_{t \in T}$ (5.1), $\{ }^8$ and one example ϕ^t was included in the respective
penev_B_F00194	6	■ 1.000	I^t	I^t	ensembles for each image I^t .
penev_B_F00195	6	■ 0.336	$\{ \phi^t(\alpha^t) \}_{t \in T}$	$\{ \phi^t(\alpha^t) \}_{t \in T}$	the objects $\{ \phi^t(\alpha^t) \}_{t \in T}$, but also all objects $\{ \phi^t(\alpha) \}_{t \in T, \alpha \in \Delta_{\alpha^t}}$, where Δ_{α^t} is
penev_B_F00196	6	■ 0.336	$\{ \phi^t(\alpha) \}_{t \in T, \alpha \in \Delta_{\alpha^t}}$	$\{ \phi^t(\alpha) \}_{t \in T, \alpha \in \Delta_{\alpha^t}}$	the objects $\{ \phi^t(\alpha^t) \}_{t \in T}$, but also all objects $\{ \phi^t(\alpha) \}_{t \in T, \alpha \in \Delta_{\alpha^t}}$, where Δ_{α^t} is
penev_B_F00197	6	■ 0.336	Δ_{α^t}	Δ_{α^t}	the objects $\{ \phi^t(\alpha^t) \}_{t \in T}$, but also all objects $\{ \phi^t(\alpha) \}_{t \in T, \alpha \in \Delta_{\alpha^t}}$, where Δ_{α^t} is
penev_B_F00198	6	■ 0.986	α^t	α^t	suitably chosen closely around α^t . In all cases, the probabilities $\mathcal{P}[\phi^t(\alpha)]$ were
penev_B_F00199	6	■ 0.986	$\mathcal{P}[\phi^t(\alpha)]$	$\mathcal{P}[\phi^t(\alpha)]$	suitably chosen closely around α^t . In all cases, the probabilities $\mathcal{P}[\phi^t(\alpha)]$ were
penev_B_F00200	6	■ 0.994	S	S	determined in the context of PCA by the projector on the PCA S -subspace $\mathbf{P}'$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00201	6	■ 0.994	$\mathbf{P}^{\prime}$	$\mathbf{P}'$	determined in the context of PCA by the projector on the PCA S -subspace $\mathbf{P}'$
penev_B_F00202	6	■ 0.966	I	I	given image I and an initial guess α_0 , calculate the entropic costs (2.49)
penev_B_F00203	6	■ 0.966	α_0	α_0	given image I and an initial guess α_0 , calculate the entropic costs (2.49)
penev_B_F00204	6	■ 0.860	$\phi(\alpha)$	$\{\phi(\alpha)\}_{\alpha \in \Delta_{\alpha_0}}$	for all images $\{\phi(\alpha)\}_{\alpha \in \Delta_{\alpha_0}}$. The maximum likelihood estimate is α that mini-
penev_B_F00205	6	■ 0.769	$S(5.4)$	$\langle S \rangle(5.4)$	mizes $\langle S \rangle$ (5.4) and, therefore, maximizes the probability $\mathcal{P}[\phi(\alpha)]$ (2.20).
penev_B_F00206	6	■ 0.769	$\mathcal{P}[\phi(\alpha)]$	$\mathcal{P}[\phi(\alpha)]$	mizes $\langle S \rangle$ (5.4) and, therefore, maximizes the probability $\mathcal{P}[\phi(\alpha)]$ (2.20).
penev_B_F00207	6	■ 0.998	α^2	α^2	the parameters of the symmetry breaking α^t . For the ensembles used in Chapter 3, the
penev_B_F00208	6	■ 0.723	$\{X_l^t, Y_l^t; X_r^t, Y_r^t\}_{t \in T}$	$\{(X_l^t, Y_l^t; X_r^t, Y_r^t)\}_{t \in T}$	$\{(X_l^t, Y_l^t; X_r^t, Y_r^t)\}_{t \in T}$. From them, the wonderful set of high-quality coefficients $\{\alpha_{G,M}^t\}_{t \in T}$
penev_B_F00209	6	■ 0.723	$\{\alpha_{G,M}^t\}_{t \in T}$	$\{\alpha_{G,M}^t\}_{t \in T}$	$\{(X_l^t, Y_l^t; X_r^t, Y_r^t)\}_{t \in T}$. From them, the wonderful set of high-quality coefficients $\{\alpha_{G,M}^t\}_{t \in T}$
penev_B_F00210	7	■ 0.861	$\langle S \rangle(N, \alpha_h^t)$	$\langle S \rangle(N, \alpha_h^t)$	The average entropic cost $\langle S \rangle(N, \alpha_h^t)$ (5.4) is plotted as a function of N for 7
penev_B_F00211	7	■ 0.998	α_h^t	α_h^t	different choices of α_h^t (5.5) for one of the images I^t in Ensemble 6 (see Ap-
penev_B_F00212	7	■ 0.999	$\phi^t(0)$	$\phi^t(0)$	The vertical axis is additionally scaled so that the in-sample example $\phi^t(0)$ (==)
penev_B_F00213	7	■ 0.985	$\langle S \rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$	$\langle S \rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$	has $\langle S \rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$ when $N \rightarrow T$, instead of the more usual, 1.
penev_B_F00214	7	■ 0.985	$N \rightarrow T$	$N \rightarrow T$	has $\langle S \rangle(N) \rightarrow \frac{\sigma_0^2}{\sum_{r=1}^N \sigma_r^2}$ when $N \rightarrow T$, instead of the more usual, 1.
penev_B_F00215	7	■ 0.999	$\phi^t(h_n)$	$\phi^t(h_n)$	The six out-of-sample examples $\phi^t(h_n)$ were produced with $h_n = 1 - s_n, s_n =$
penev_B_F00216	7	■ 0.999	$h_n = 1 - s_n, s_n =$	$h_n = 1 - s_n, s_n =$	The six out-of-sample examples $\phi^t(h_n)$ were produced with $h_n = 1 - s_n, s_n =$
penev_B_F00217	7	■ 1.000	$1.05^{n/2}$	$1.05^{n/2}$	$1.05^{n/2}$, for $n \in \{-3, -2, -1, 1, 2, 3\}$, as outlined in Appendix A.1.
penev_B_F00218	7	■ 1.000	$n \in \{-3, -2, -1, 1, 2, 3\}$	$n \in \{-3, -2, -1, 1, 2, 3\}$	$1.05^{n/2}$, for $n \in \{-3, -2, -1, 1, 2, 3\}$, as outlined in Appendix A.1.
penev_B_F00219	7	■ 0.998	I^t, α^t	I^t, α^t	for definition) are shown on Fig. 5.5. For all 1216 images I^t , α^t was determined.
penev_B_F00220	7	■ 1.000	$\phi^t(h) = H(I^t; \alpha_h^t)$	$\phi^t(h) = H(I^t; \alpha_h^t)$	From each α^t , several examples $\phi^t(h) = H(I^t; \alpha_h^t)$ were produced with
penev_B_F00221	7	■ 1.000	$\phi^t(0)$	$\phi^t(0)$	as outlined in Appendix A.1. Only the examples $\phi^t(0)$ were used to derive
penev_B_F00222	7	■ 0.997	$T=1216$	$T=1216$	the modes, hence $T=1216$.
penev_B_F00223	7	■ 1.000	$\langle S \rangle(N) \rightarrow 1$	$\langle S \rangle(N) \rightarrow 1$	Two effects can be observed. First, the in-sample effect causes $\langle S \rangle(N) \rightarrow 1$
penev_B_F00224	7	■ 1.000	$h=0$	$h=0$	with $N \rightarrow T$ for $h=0$. Also, for smaller deviations from in-sample-ness $ h $,
penev_B_F00225	7	■ 1.000	$ h $	$ h $	with $N \rightarrow T$ for $h=0$. Also, for smaller deviations from in-sample-ness $ h $,
penev_B_F00226	7	■ 0.509	$\langle S \rangle(T, \alpha_h)$	$\langle S \rangle(T, \alpha_h)$	the entropic cost of the un-truncated representation, $\langle S \rangle(T, \alpha_h)$, is generally
penev_B_F00227	7	■ 1.000	$N \in [350, 750]$	$N \in [350, 750]$	Second, for the “reasonable regime” $N \in [350, 750]$, the out-of-sample ex-
penev_B_F00228	7	■ 0.999	$\phi^t(h_{+1})$	$\phi^t(h_{+1})$	ample $\phi^t(h_{+1})$ is more probable than the in-sample example $\phi^t(0)$. Therefore,
penev_B_F00229	7	■ 0.999	$\phi^t(0)$	$\phi^t(0)$	ample $\phi^t(h_{+1})$ is more probable than the in-sample example $\phi^t(0)$. Therefore,
penev_B_F00230	7	■ 0.922	$S - \log N$	$S - \log N$	Figure 5.6: $S - \log N$ Diagrams and Scale Symmetry Breaking (+)
penev_B_F00231	7	■ 1.000	$n \in \{-4, -3, 3, 4\}$	$n \in \{-4, -3, 3, 4\}$	the examples $\phi^t(h_n)$, with $n \in \{-4, -3, 3, 4\}$, have been substituted with the
penev_B_F00232	7	■ 0.996	$\phi^t(h_m)$	$\phi^t(h_m)$	examples $\phi^t(h_m)$ from Ensemble 7 (see Appendix A.1 for definition) with $h_m =$
penev_B_F00233	7	■ 0.996	$h_m =$	$h_m =$	examples $\phi^t(h_m)$ from Ensemble 7 (see Appendix A.1 for definition) with $h_m =$
penev_B_F00234	7	■ 0.990	$1 - s_m, s_m = 1.025^{m/3}$	$1 - s_m, s_m = 1.025^{m/3}$	$1 - s_m, s_m = 1.025^{m/3}$. In this notation, $n=1$ produces almost the same
penev_B_F00235	7	■ 0.990	$n=1$	$n=1$	$1 - s_m, s_m = 1.025^{m/3}$. In this notation, $n=1$ produces almost the same
penev_B_F00236	7	■ 1.000	$m=3$	$m=3$	example as $m=3$.
penev_B_F00237	7	■ 0.911	$(-); -1(-); +1(+)$	$(-); -1(-); +1(+)$	The values of m are: -2 , labeled with $(-)$; -1 $(-)$; $+1$ $(+)$; and $+2$ $(+)$.
penev_B_F00238	7	■ 0.911	$+2(++)$	$+2(++)$	The values of m are: -2 , labeled with $(-)$; -1 $(-)$; $+1$ $(+)$; and $+2$ $(+)$.
penev_B_F00239	7	■ 0.813	α_{+1}^t	α_{+1}^t	for a reasonable truncation, α_{+1}^t is a better parameter of symmetry breaking
penev_B_F00240	7	■ 1.000	α^t	α^t	than α^t .
penev_B_F00241	7	■ 1.000	h	h	optimal value for h .
penev_B_F00242	7	■ 0.990	$N, h_{\text{best}}(N) > 0$	$N, h_{\text{best}}(N) > 0$	mensionalities $N, h_{\text{best}}(N) > 0$. Also, very big values of $ h $ result in very
penev_B_F00243	7	■ 0.963	$\langle S \rangle(N, \alpha_h^t)$	$\langle S \rangle(N, \alpha_h^t)$	Evidently, for all N , $\langle S \rangle(N, \alpha_h^t)$ has a local minimum for $h > 0$. As N
penev_B_F00244	7	■ 0.963	$h > 0$	$h > 0$	Evidently, for all N , $\langle S \rangle(N, \alpha_h^t)$ has a local minimum for $h > 0$. As N
penev_B_F00245	7	■ 1.000	$h \approx +1.5\%$	$h \approx +1.5\%$	increases, this minimum is achieved for $h \approx +1.5\%$, and is forced to $h=0$ as
penev_B_F00246	7	■ 0.998	$h=1.5\%$	$h=1.5\%$	the in-sample effect takes over. Therefore, one can conclude that $h=1.5\%$
penev_B_F00247	7	■ 1.000	$T > N$	$T > N$	which will result in $T > N$, there will be one optimal value of h for all N . A
penev_B_F00248	7	■ 0.777	$\phi^t(h)$	$\phi^t(h)$	studied $\phi^t(h)$. In this manner, for each image I^t , we produced the <i>trajectory</i>
penev_B_F00249	7	■ 1.000	$\{\Phi^{*+} \phi^t(h)\} \subset S$	$\{\Phi^{*+} \phi^t(h)\} \subset S$	$\{\Phi^{*+} \phi^t(h)\} \subset S$.
penev_B_F00250	7	■ 0.998	$\ \Phi^{*+} \phi^t(h)\ _S^2$	$\ \Phi^{*+} \phi^t(h)\ _S^2$	jectory and the origin, $\ \Phi^{*+} \phi^t(h)\ _S^2$ as the entropy (2.20) of the example $\phi^t(h)$
penev_B_F00251	7	■ 0.998	$\phi^t(h)$	$\phi^t(h)$	jectory and the origin, $\ \Phi^{*+} \phi^t(h)\ _S^2$ as the entropy (2.20) of the example $\phi^t(h)$
penev_B_F00252	7	■ 1.000	k	k	In general, α has k , more than one, independent components for the
penev_B_F00253	7	■ 0.937	$\phi^t(\alpha)$	$\phi^t(\alpha)$	special case discussed in Section 5.1 they are four (5.1)—and $\phi^t(\alpha)$ is a k -
penev_B_F00254	7	■ 0.941	$\Phi^{*+} \phi(\alpha)$	$\Phi^{*+} \phi(\alpha)$	points on the manifold $\Phi^{*+} \phi(\alpha)$ is a function of $\alpha - S(\alpha)$. On one hand, for
penev_B_F00255	7	■ 0.941	$\alpha - S(\alpha)$	$\alpha - S(\alpha)$	points on the manifold $\Phi^{*+} \phi(\alpha)$ is a function of $\alpha - S(\alpha)$. On one hand, for

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00256	7	■ 1.000	$\phi(\alpha)$	$\phi(\alpha)$	values of α that are very wrong, the probability that $\phi(\alpha)$ is in the ensemble
penev_B_F00257	7	■ 0.996	$S(\alpha)$	$S(\alpha)$	will be very low and $S(\alpha)$ will be very large a tree is not a face, as well as
penev_B_F00258	7	■ 0.995	$S_{\text{\texttt{max}}}$	$S_{\max}$	one can find a reasonable threshold $S_{\max}$ and a finite compact region on the
penev_B_F00259	7	■ 0.904	Δ	Δ	mauifold Δ , such that $\{\alpha S(\alpha) < S_{\max}\} \subset \Delta$.
penev_B_F00260	7	■ 0.904	$\left\{ \alpha \mid S(\alpha) < S_{\max} \right\} \subset \Delta$	$\{\alpha \mid S(\alpha) < S_{\max}\} \subset \Delta$	mauifold Δ , such that $\{\alpha S(\alpha) < S_{\max}\} \subset \Delta$.
penev_B_F00261	8	■ 0.787	α_{GM}^t	α_{GM}^t	and for a reasonable choice for N , and for a reasonable initial estimate $\alpha_{G,M}^t$.
penev_B_F00262	8	■ 1.000	$N \approx 10$	$N \approx 10$	Suppose that the knowledge extends only up to $N \approx 10$. In that context, only
penev_B_F00263	8	■ 0.791	X, Y	X, Y	contact, the example is centered at the fovea and, therefore, the (X, Y) part
penev_B_F00264	8	■ 1.000	Ψ_3	Ψ_3	due to the strong “Lambertian” mode Ψ_3 (Fig. 3.9) and its symmetrizations.
penev_B_F00265	8	■ 0.893	Ψ_2, Ψ_4	Ψ_2, Ψ_4	that other strong irrelevant modes—the background modes Ψ_2, Ψ_4 , and Ψ_δ ,
penev_B_F00266	8	■ 0.893	Ψ_5	Ψ_5	that other strong irrelevant modes—the background modes Ψ_2, Ψ_4 , and Ψ_δ ,
penev_B_F00267	8	■ 0.993	Ψ_1^{12}	Ψ_1^{12}	will suffice—a very frugal definition of Ψ_1 . ¹²
penev_B_F00268	8	■ 1.000	$\alpha, \mathcal{P}_\alpha$	$\alpha, \mathcal{P}_\alpha$	$\mathcal{P}_{objects}$, and the symmetric density of the parameter of the breaking $\alpha, \mathcal{P}_\alpha$.
penev_B_F00269	8	■ 0.988	$\beta_{\text{\texttt{feature}}}$	$\beta_{\text{\texttt{feature}}}$	$\beta_{\text{\texttt{feature}}}$, as a parameter for the breaking of the local symmetry. Analogously,
penev_B_F00270	8	■ 0.994	$\mathcal{P}_{\text{\texttt{feature}}}$	$\mathcal{P}_{\text{\texttt{feature}}}$	ensemble, this time of features, and also, its probability density $\mathcal{P}_{\text{\texttt{feature}}}$, along
penev_B_F00271	8	■ 1.000	$\mathcal{P}_\beta$	$\mathcal{P}_\beta$	with the probability density $\mathcal{P}_\beta$, as a parameterization of $\mathcal{P}_{objects}$.
penev_B_F00272	8	■ 1.000	Ψ_r	Ψ_r	at the fovea; this obviates the need for any genetic storage of Ψ_r .
penev_B_F00273	8	■ 0.429	$\phi^t(\mathbf{x})$	$\phi^t(\mathbf{x})$	receptors” — $\phi^t(\mathbf{x})$ was interpreted as the luminosity value at the grid point $\mathbf{x}$. This worked
penev_B_F00274	8	■ 1.000	A	A	weak assumptions, if area A projects forward to area B , then almost always
penev_B_F00275	8	■ 1.000	B	B	weak assumptions, if area A projects forward to area B , then almost always
penev_B_F00276	8	■ 0.932	20%	20%	Ensemble 6, where about 20% of the faces have eyeglasses, the “cheek-bone”
penev_B_F00277	8	■ 0.996	$P_m(\mathbf{x})$	$P_m(\mathbf{x})$	Unfortunately, its reconstructor $P_m(\mathbf{x})$ (Fig. 5.4e) will tend to create an illusion
penev_B_F00278	8	■ 0.662	$\phi_{100}^{\text{err}} - \phi_{500}^{\text{err}}$	$\phi_{100}^{\text{err}} - \phi_{500}^{\text{err}}$	of faint eyeglasses (cf. $\phi_{100}^{\text{err}} - \phi_{500}^{\text{err}}$ on Fig. 5.2), ¹⁵ and it will take many more
penev_B_F00279	8	■ 0.919	$\left\{ \alpha_{G.M.}^t \right\}_{t \in T}$	$\{\alpha_{G.M.}^t\}_{t \in T}$	well for two reasons: we had a good initial guess $\{\alpha_{G.M.}^t\}_{t \in T}$ about the symmetry breaking
penev_B_F00280	8	■ 0.769	$\mathcal{P}_{\text{\texttt{abjects}}}$	$\mathcal{P}_{\text{\texttt{abjects}}}$	$\mathcal{P}_{objects}$, probability distributions in one representational module (Fig. 5.1). In general, when
penev_B_F00281	8	■ 0.916	A_3	A_3	the forward, or <i>ascending</i> , pathway originate in the superficial pyramidal cells of area A ,
penev_B_F00282	9	■ 0.985	$O^{\text{err}}(\mathbf{x})$	$O^{\text{err}}(\mathbf{x})$	ule will then calculate $O^{\text{err}}(\mathbf{x})$ and will feed it to the cheekbone module for
penev_B_F00283	9	■ 1.000	U	U	achieved by topography and the simplistic fixing of the U -symmetry (4.16); as long as we
penev_B_F00284	9	■ 0.996	$\mathcal{M} \subset V$	$\mathcal{M} \subset V$	values in a sparse subset $\mathcal{M} \subset V$ of the receptor grid $\{\mathbf{x}\}_{\mathbf{x} \in V}$ (Fig. 4.6). An
penev_B_F00285	9	■ 0.996	$\{\mathbf{x}\}_{\mathbf{x} \in V}$	$\{\mathbf{x}\}_{\mathbf{x} \in V}$	values in a sparse subset $\mathcal{M} \subset V$ of the receptor grid $\{\mathbf{x}\}_{\mathbf{x} \in V}$ (Fig. 4.6). An
penev_B_F00286	9	■ 0.868	$\mathbf{P} = \mathbf{O}^* \mathbf{O} (4.28)$	$\mathbf{P} = \mathbf{O}^* \mathbf{O} (4.28)$	residual correlations $\mathbf{P} = \mathbf{O}^* \mathbf{O} (4.28)$, and, in the symmetric case, their support
penev_B_F00287	9	■ 0.949	$\mathbf{O} * \mathbf{O} = \mathbf{1} (4.12)$	$\mathbf{O} * \mathbf{O} = \mathbf{1} (4.12)$	collapses to the point in the center, $\mathbf{O}^* \mathbf{O} = \mathbf{1} (4.12)$. Therefore, the knowledge
penev_B_F00288	9	■ 0.536	$\left\{ O(\mathbf{x}_m) \right\}_{\mathbf{x}_m \in \mathcal{M}}$	$\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$	of the values $\{O(\mathbf{x}_m)\}_{\mathbf{x}_m \in \mathcal{M}}$ does not provide any information about the rest
penev_B_F00289	9	■ 0.961	$\mathbf{f} \in \Delta_j$	$\mathbf{f} \in \Delta_j$	$\mathbf{f} \in \Delta_j$. Therefore, the correlation function of the sensory ensemble Φ_j , as seen
penev_B_F00290	9	■ 0.961	Φ_j	Φ_j	$\mathbf{f} \in \Delta_j$. Therefore, the correlation function of the sensory ensemble Φ_j , as seen
penev_B_F00291	9	■ 0.994	j	j	by a neuron in the j -th frequency band, is $\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$ with (cf. 4.27)
penev_B_F00292	9	■ 0.994	$\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$	$\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$	by a neuron in the j -th frequency band, is $\mathbf{R}_j \equiv \mathbf{K}_j^{(-2)}$ with (cf. 4.27)
penev_B_F00293	9	■ 0.692	$\phi_j = \mathbf{P}_j \phi$	$\phi_j = \mathbf{P}_j \phi$	The sensory signal that a multi-scale neuron sees is $\phi_j = \mathbf{P}_j \phi$ with $\mathbf{P}_j = \mathbf{K}_j^{(0)}$,
penev_B_F00294	9	■ 0.692	$\mathbf{P}_j = \mathbf{K}_j^{(0)}$	$\mathbf{P}_j = \mathbf{K}_j^{(0)}$	The sensory signal that a multi-scale neuron sees is $\phi_j = \mathbf{P}_j \phi$ with $\mathbf{P}_j = \mathbf{K}_j^{(0)}$,
penev_B_F00295	9	■ 0.910	$\mathbf{K}_j = \mathbf{K}_j^{(1)}$	$\mathbf{K}_j = \mathbf{K}_j^{(1)}$	Now, the outputs of the action of $\mathbf{K}_j = \mathbf{K}_j^{(1)}$ on ϕ_j is ²⁰
penev_B_F00296	9	■ 0.910	ϕ_j	ϕ_j	Now, the outputs of the action of $\mathbf{K}_j = \mathbf{K}_j^{(1)}$ on ϕ_j is ²⁰
penev_B_F00297	9	■ 0.499	$O(\mathbf{x})$	$O(\mathbf{x})$	²⁰ Notably, j is not a substitute for the receptor grid index $\mathbf{x}$ in $O(\mathbf{x})$; there are several band-
penev_B_F00298	9	■ 1.000	O_j	O_j	passed dense LFA outputs O_j , which collectively carry the information that was previously
penev_B_F00299	9	■ 1.000	O	O	carried entirely by O . When we want to emphasize their dependence on $\{\mathbf{x}\}$, we will write
penev_B_F00300	9	■ 1.000	$\{\mathbf{x}\}$	$\{\mathbf{x}\}$	carried entirely by O . When we want to emphasize their dependence on $\{\mathbf{x}\}$, we will write
penev_B_F00301	9	■ 0.439	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$. For the ensemble of monaural sound, $\mathbf{x}$ is the time index t ; for the static
penev_B_F00302	9	■ 0.439	t	t	$(\hat{\mathbf{x}}, O_j) = O_j(\mathbf{x})$. For the ensemble of monaural sound, $\mathbf{x}$ is the time index t ; for the static
penev_B_F00303	9	■ 0.999	(x, y)	(x, y)	natural images, the pair (x, y) ; for the dynamic natural “movies,” the triplet (x, y, t) .
penev_B_F00304	9	■ 0.999	(x, y, t)	(x, y, t)	natural images, the pair (x, y) ; for the dynamic natural “movies,” the triplet (x, y, t) .
penev_B_F00305	9	■ 1.000	$\{\Delta_j\}$	$\{\Delta_j\}$	²¹ Here, we are not addressing the possibility that the bands $\{\Delta_j\}$ may happen to be non-
penev_B_F00306	9	■ 0.999	$\mathbf{O}_j^* \mathbf{O}_k$	$\mathbf{O}_j^* \mathbf{O}_k$	question what $\mathbf{O}_j^* \mathbf{O}_k$ is for $j \neq k$.
penev_B_F00307	9	■ 0.999	$j \neq k$	$j \neq k$	question what $\mathbf{O}_j^* \mathbf{O}_k$ is for $j \neq k$.
penev_B_F00308	9	■ 1.000	F_r	F_r	One way to deal uniformly with all cases is through the “noise suppression” factor F_r .
penev_B_F00309	9	■ 0.995	$(\mathbf{P} = \mathbf{1})$	$(\mathbf{P} = \mathbf{1})$	page 61); it could be decorrelated fully $(\mathbf{P} = \mathbf{1})$ by the retinal filters (4.26), but
penev_B_F00310	9	■ 1.000	$\{\phi_j\}$	$\{\phi_j\}$	among the bands $\{\Delta_j\}$, resulting in $\{\phi_j\}$, so is the dense LFA output, $\{O_j\}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00311	9	■ 1.000	$\left\{0_{\{j\}}\right\}$	$\{O_j\}$	among the bands $\{\hat{\Delta}_j\}$, resulting in $\{\phi_j\}$, so is the dense LFA output, $\{O_j\}$.
penev_B_F00312	9	■ 1.000	$\mathrm{0}_{\{j\}}$	O_j	the respective output ensembles O_j , are symmetric. The multi-scale process-
penev_B_F00313	9	■ 1.000	$0_{\{3\}}$	O_3	after a specific example ϕ has been presented and the degenerate outputs O_j
penev_B_F00314	9	■ 0.858	$\mathrm{K}_{\{j\}}^{\{n\}}=\sum_{r\in\Delta_j}\psi_r\left(\frac{1}{\sigma_r}\right)^n(\psi_r,)$	$\mathbf{K}_j^{(n)} = \sum_{r \in \Delta_j} \psi_r \left(\frac{1}{\sigma_r} \right)^n (\psi_r,)$	representation of the type $\mathbf{K}_j^{(n)} = \sum_{r \in \Delta_j} \psi_r \left(\frac{1}{\sigma_r} \right)^n (\psi_r,)$ (cf. 4.30) can result in additional
penev_B_F00315	9	■ 0.695	$\{ \}^{\{23\}} \mathrm{As}$	$^{23}\mathrm{As}$	$^{23}\mathrm{As}$ with any symmetry breaking, it can be understood as a definition of a restricted
penev_B_F00316	10	■ 0.723	$\left\{ \mathbf{a}_{\{m\}}(\mathbf{x}) \right\}_{\mathbf{x}_m \in \mathcal{M}} \in \mathcal{M}$ (4.68)	$\{a_m(\mathbf{x})\}_{x_m \in \mathcal{M}}$ (4.68)	struction as either the set of interpolation reconstructors $\{a_m(\mathbf{x})\}_{\mathbf{x}_m \in \mathcal{M}}$ (4.68)
penev_B_F00317	10	■ 0.869	$P_{\{m\}}(\mathbf{x})$ (4.73)	$P_m(\mathbf{x})$ (4.73)	$P_m(\mathbf{x})$ (4.73) .
penev_B_F00318	10	■ 1.000	$ \mathcal{M} =M-1$	$ \mathcal{M} = M - 1$	We will consider the case when $ \mathcal{M} = M - 1$ units are currently active and the
penev_B_F00319	10	■ 1.000	M	M	M -th one is being activated. ²⁷ Then, two things happen in the reconstruction
penev_B_F00320	10	■ 1.000	$M-1$	$M - 1$	generated; and corrections to the old $M-1$ reconstructors are made.
penev_B_F00321	10	■ 0.403	$\mathbf{Q}$ (4.63)	$\mathbf{Q}$ (4.63)	inversion of the matrix $\mathbf{Q}$ (4.63) and the correlators $P_m(\mathbf{x})$ (4.68); the latter
penev_B_F00322	10	■ 0.403	$P_{\{m\}}(\mathbf{x})$ (4.68)	$P_m(\mathbf{x})$ (4.68)	inversion of the matrix $\mathbf{Q}$ (4.63) and the correlators $P_m(\mathbf{x})$ (4.68); the latter
penev_B_F00323	10	■ 0.999	$n=n^{\prime}+n^{\prime\prime}$	$n = n' + n''$	In general, for $n = n' + n''$, any $n \times n$ matrix W can be partitioned as
penev_B_F00324	10	■ 0.999	$n \times n$	$n \times n$	In general, for $n = n' + n''$, any $n \times n$ matrix W can be partitioned as
penev_B_F00325	10	■ 0.999	W	W	In general, for $n = n' + n''$, any $n \times n$ matrix W can be partitioned as
penev_B_F00326	10	■ 0.745	$\mathbf{1}_{n^{\prime}}$	$\mathbf{1}_{n'}$	where $\mathbf{1}_{n'}$ is the $n' \times n'$ unit matrix; analogously for $\mathbf{1}_{n''}$; and the partition-
penev_B_F00327	10	■ 0.745	$n^{\prime} \times n^{\prime}$	$n' \times n'$	where $\mathbf{1}_{n'}$ is the $n' \times n'$ unit matrix; analogously for $\mathbf{1}_{n''}$; and the partition-
penev_B_F00328	10	■ 0.745	$\mathbf{1}_{n^{\prime\prime}}$	$\mathbf{1}_{n''}$	where $\mathbf{1}_{n'}$ is the $n' \times n'$ unit matrix; analogously for $\mathbf{1}_{n''}$; and the partition-
penev_B_F00329	10	■ 1.000	W^{-1}	W^{-1}	derstanding that they are as in (5.9). Then, for the inverse W^{-1} , we have
penev_B_F00330	10	■ 0.999	w^{-1}	w^{-1}	Thus, W^{-1} is calculated by a correction to w^{-1} , and only the quantities x , y ,
penev_B_F00331	10	■ 0.999	x, y	x, y	Thus, W^{-1} is calculated by a correction to w^{-1} , and only the quantities x , y ,
penev_B_F00332	10	■ 1.000	z	z	and z are needed additionally; W and w are never needed explicitly.
penev_B_F00333	10	■ 1.000	w	w	and z are needed additionally; W and w are never needed explicitly.
penev_B_F00334	10	■ 1.000	$M \times M$	$M \times M$	After the M -th unit becomes active, we identify W with the $M \times M$ ma-
penev_B_F00335	10	■ 0.513	$(n=M)$	$(n = M)$	trix $\mathbf{Q}$ $[n = M]$, whose matrix elements are (4.63)
penev_B_F00336	10	■ 0.933	$(M-1) \times (M-1)$	$(M - 1) \times (M - 1)$	Also, we identify w with the $[M-1] \times [M-1]$ matrix from the previous step,
penev_B_F00337	10	■ 0.861	$w \equiv q$	$w \equiv q$	$w \equiv q$, comprised by the first $M - 1$ rows and columns of $\mathbf{Q}$, $q_{ml} = Q_{ml} _{l,m=1}^{M-1}$.
penev_B_F00338	10	■ 0.861	$\mathbf{Q}, q_{ml} = Q_{ml} _{l,ns=1}^{M-1}$	$\mathbf{Q}, q_{ml} = Q_{ml} _{l,ns=1}^{M-1}$	$w \equiv q$, comprised by the first $M - 1$ rows and columns of $\mathbf{Q}$, $q_{ml} = Q_{ml} _{l,m=1}^{M-1}$.
penev_B_F00339	10	■ 0.994	$w, n^{\prime}=M-1$	$w, n' = M - 1$	With these W and w , $n' = M - 1$, hence $n'' = 1$; then x is a $(M - 1)$ -
penev_B_F00340	10	■ 0.994	$n^{\prime\prime}=1$	$n'' = 1$	With these W and w , $n' = M - 1$, hence $n'' = 1$; then x is a $(M - 1)$ -
penev_B_F00341	10	■ 0.994	x	x	With these W and w , $n' = M - 1$, hence $n'' = 1$; then x is a $(M - 1)$ -
penev_B_F00342	10	■ 0.994	$(M-1)$	$(M - 1)$	With these W and w , $n' = M - 1$, hence $n'' = 1$; then x is a $(M - 1)$ -
penev_B_F00343	10	■ 0.908	y	y	dimensional column vector; y , a row vector; z and $\bar{z}$, scalars. Because $\mathbf{P}$ (4.29)
penev_B_F00344	10	■ 0.908	$\bar{z}$	$\bar{z}$	dimensional column vector; y , a row vector; z and $\bar{z}$, scalars. Because $\mathbf{P}$ (4.29)
penev_B_F00345	10	■ 0.806	p	p	and, therefore, $\mathbf{Q}$ are symmetric, we can define p as the row vector ²⁸
penev_B_F00346	10	■ 0.917	$\tilde{z}$	$\tilde{z}$	Because $\mathbf{Q}$ is symmetric and $\tilde{z}$ is a scalar, the second term in (5.11) factorizes
penev_B_F00347	10	■ 0.870	$O^{\mathrm{rec}}(\mathbf{x})$	$O^{\mathrm{rec}}(\mathbf{x})$	means that $O^{\mathrm{rec}}(\mathbf{x})$ is a V -dimensional column vector, which is a linear com-
penev_B_F00348	10	■ 1.000	$A^{\mathrm{m}}(\mathbf{x})$	$A^m(\mathbf{x})$	bination of the M (uppercase) column vectors $A^m(\mathbf{x})$ with the M coefficients
penev_B_F00349	10	■ 1.000	$O_{\{m\}}=O\left(\mathbf{x}_{\{m\}}\right)$	$O_m = O(\mathbf{x}_m)$	$O_m = O(\mathbf{x}_m)$. Analogously, the column vector $A^m(\mathbf{x})$ for the m -th reconstruc-
penev_B_F00350	10	■ 1.000	$\{ \}^{\{28\}} p^{\mathrm{T}}$	$^{28}p^T$	$^{28}p^T$ is a column vector, which is the <i>transpose</i> of the row vector p .
penev_B_F00351	10	■ 1.000	$P^{\mathrm{l}}(\mathbf{x})$	$P^l(\mathbf{x})$	is a linear combination of the M correlators $P^l(\mathbf{x})$, centered at the active grid
penev_B_F00352	10	■ 0.967	$\mathbf{x}_{\mathrm{l}}$	$\mathbf{x}_l$	points, $\mathbf{x}_l$, with the coefficients $(\mathbf{Q}^{-1})_l^m$, from the m -th column of $\mathbf{Q}^{-1}$.
penev_B_F00353	10	■ 0.967	$\left(\mathbf{Q}^{-1}\right)_{\mathrm{l}}^m$	$(\mathbf{Q}^{-1})_l^m$	points, $\mathbf{x}_l$, with the coefficients $(\mathbf{Q}^{-1})_l^m$, from the m -th column of $\mathbf{Q}^{-1}$.
penev_B_F00354	10	■ 0.994	$\left[a^{\mathrm{m}}(\mathbf{x}) \quad 0\right]$	$[a^m(\mathbf{x}) \quad 0]$	Thus, the new reconstructors $A^m(\mathbf{x})$ are the old ones, $\begin{bmatrix} a^m(\mathbf{x}) & 0 \end{bmatrix}$, with a
penev_B_F00355	10	■ 0.880	$A(\mathbf{x})$	$A(\mathbf{x})$	whose entire $\mathbf{x}$ -dependence is given by $A(\mathbf{x})$. Indeed, the new reconstructor
penev_B_F00356	10	■ 1.000	$A^{\mathrm{M}}(\mathbf{x})$	$A^M(\mathbf{x})$	$A^M(\mathbf{x})$ is proportional to $A(\mathbf{x})$ with a coefficient $\tilde{z}$,
penev_B_F00357	10	■ 1.000	$\tilde{z}_1$	$\tilde{z}_1$	$A^M(\mathbf{x})$ is proportional to $A(\mathbf{x})$ with a coefficient $\tilde{z}$,
penev_B_F00358	10	■ 0.886	$a^{\mathrm{m}}(\mathbf{x})$	$a^m(\mathbf{x})$	and the correction to any old reconstructor $a^m(\mathbf{x})$, is proportional to $A^M(\mathbf{x})$
penev_B_F00359	10	■ 0.886	$A^{\mathrm{M}}(\mathbf{x})$	$A^M(\mathbf{x})$	and the correction to any old reconstructor $a^m(\mathbf{x})$, is proportional to $A^M(\mathbf{x})$
penev_B_F00360	10	■ 1.000	$-u^{\mathrm{m}}$	$-u^m$	with a coefficient $-u^m$:
penev_B_F00361	10	■ 0.981	$u=pq^{-1}$	$u = pq^{-1}$	Now, we are well motivated to inquire into the meaning of $u = pq^{-1}$ (5.15).
penev_B_F00362	10	■ 0.993	u^{m}	u^m	From (5.13) (cf. 5.18), u^m is the old reconstructor $a^m(\mathbf{x})$, evaluated at the
penev_B_F00363	10	■ 0.679	$\mathbf{x}_{\mathrm{M}}$	$\mathbf{x}_M$	newly active point, $\mathbf{x}_M$,

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penev_B_F00364	10	0.815	$p^{\{l\}}(\mathbf{u}) \quad u^{\{T\}}=p^{\{l\}}(\mathbf{x})\left(q^{-1}\right)^m P^M\left(\mathbf{x}_m\right)=a^m(\mathbf{x}) P^M\left(\mathbf{x}_m\right)$		Also, (cf. 5.20), $p^{\{l\}}(\mathbf{x}) u^{\{T\}}=p^{\{l\}}(\mathbf{x})\left(q^{-1}\right)^m P^M\left(\mathbf{x}_m\right)=a^m(\mathbf{x}) P^M\left(\mathbf{x}_m\right)$; hence
penev_B_F00365	10	0.943	$a^{\{m\}}(\mathbf{x}) \quad P^{\{M\}}\left(\mathbf{x}_m\right)$	$a^m(\mathbf{x}) P^M\left(\mathbf{x}_m\right)$	The last term in (5.25), $a^m(\mathbf{x}) P^M\left(\mathbf{x}_m\right)$, is analogous to (5.17) with the
penev_B_F00366	10	0.629	$\left(\mathbf{x}_m\right)$	$\left\{\mathbf{x}_m\right\}$	evaluated at $\left\{\mathbf{x}_m\right\}$ is $P^M(\mathbf{x})$, rather than $O(\mathbf{x})$. Therefore, it is useful, given
penev_B_F00367	10	0.629	$P^{\{M\}}(\mathbf{x})$	$P^M(\mathbf{x})$	evaluated at $\left\{\mathbf{x}_m\right\}$ is $P^M(\mathbf{x})$, rather than $O(\mathbf{x})$. Therefore, it is useful, given
penev_B_F00368	10	1.000	$V(\mathbf{x})$	$V(\mathbf{x})$	the current restriction $\mathcal{M}$, to define, for any vector $V(\mathbf{x})$, the <i>reconstruction</i>
penev_B_F00369	11	0.926	$P^{\{M\}}(\mathbf{x})$	$P^M(\mathbf{x})$	Remarkably, $A(\mathbf{x})$ is the difference between the true value of $P^M(\mathbf{x})$ and its
penev_B_F00370	11	1.000	$\bar{z}^{-1}=\bar{z}-u p^T$	$\bar{z}^{-1}=\bar{z}-u p^T$	$\bar{z}^{-1}=\bar{z}-u p^T$, and from (5.14), (5.24), (5.13), and (5.29),
penev_B_F00371	11	0.868	$a^{\{m\}}(\mathbf{x})$	$a^m(\mathbf{x})$	predictors, $a^m(\mathbf{x})$, are corrected by a negative feedback of this error, in propor-
penev_B_F00372	11	0.870	$a^{\{m\}}\left(\mathbf{x}_M\right)$	$a^m\left(\mathbf{x}_M\right)$	tionality with their original prediction strengths, $a^m\left(\mathbf{x}_M\right)$.
penev_B_F00373	11	0.582	$O^{\text {rec }}(\mathbf{x})=\mathcal{R}_{O}^{M-1}(\mathbf{x})$	$O^{\text {rec }}(\mathbf{x})=\mathcal{R}_O^{M-1}(\mathbf{x})$	the current estimate $O^{\text {rec }}(\mathbf{x})=\mathcal{R}_O^{M-1}(\mathbf{x})$, the current error $O^{\text {err }}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$
penev_B_F00374	11	0.582	$O^{\text {err }}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$	$O^{\text {err }}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$	the current estimate $O^{\text {rec }}(\mathbf{x})=\mathcal{R}_O^{M-1}(\mathbf{x})$, the current error $O^{\text {err }}(\mathbf{x})=\mathcal{D}_O(\mathbf{x})$
penev_B_F00375	11	0.381	$\Delta O^{\text {rec }}(\mathbf{x})$	$\Delta O^{\text {rec }}(\mathbf{x})$	introduces a correction $\Delta O^{\text {rec }}(\mathbf{x})$, which is exactly equal to the error at that
penev_B_F00376	11	0.716	$\mathcal{D}_O\left(\mathbf{x}_M\right)$	$\mathcal{D}_O\left(\mathbf{x}_M\right)$	on the current reconstruction error, $\mathcal{D}_O\left(\mathbf{x}_M\right)$.
penev_B_F00377	11	0.490	$O^{\text {err }}\left(\mathbf{x}_M\right)=\mathcal{D}_O\left(\mathbf{x}_M\right)$	$O^{\text {err }}\left(\mathbf{x}_M\right)=\mathcal{D}_O\left(\mathbf{x}_M\right)$	$O^{\text {err }}\left(\mathbf{x}_M\right)=\mathcal{D}_O\left(\mathbf{x}_M\right)$.
penev_B_F00378	11	1.000	$f(t)$	$f(t)$	like to understand it as an infinite source $f(t)$ (cf. Appendix A.6), from which
penev_B_F00379	11	0.551	$\phi^{t_n}\left(\mathbf{x}_k\right)=f\left(t_{n-k}\right)$	$\phi^{t_n}\left(\mathbf{x}_k\right)=f\left(t_{n-k}\right)$	we will construct finite-dimensional ensembles of examples $\phi^{t_n}\left(\mathbf{x}_k\right)=f\left(t_{n-k}\right)$
penev_B_F00380	11	1.000	t_n	t_n	by imposing a finite discrete window of size V around the time t_n (A.13).
penev_B_F00381	11	0.975	$\left\{f_{\chi}\right\}_{\chi \in X}$	$\left\{f_{\chi}\right\}_{\chi \in X}$	The <i>Encoder</i> (α) has access to the signal $\left\{f_{\chi}\right\}_{\chi \in X}$. Based on the part of the
penev_B_F00382	11	0.467	$\mathbf{x}_n$	$\mathbf{x}_n$	location $\mathbf{x}_n$, an estimate $f^{\text {rec }}\left(\mathbf{x}_n\right) \equiv f_n^{\text {rec }}$, and the <i>Quantizer</i> quantizes it to $\hat{f}_n$.
penev_B_F00383	11	0.467	$f^{\text {rec }}\left(\mathbf{x}_n\right) \equiv f_n^{\text {rec }}$	$f^{\text {rec }}\left(\mathbf{x}_n\right) \equiv f_n^{\text {rec }}$	location $\mathbf{x}_n$, an estimate $f^{\text {rec }}\left(\mathbf{x}_n\right) \equiv f_n^{\text {rec }}$, and the <i>Quantizer</i> quantizes it to $\hat{f}_n$.
penev_B_F00384	11	0.467	f_n	f_n	location $\mathbf{x}_n$, an estimate $f^{\text {rec }}\left(\mathbf{x}_n\right) \equiv f_n^{\text {rec }}$, and the <i>Quantizer</i> quantizes it to $\hat{f}_n$.
penev_B_F00385	11	0.996	$e_n=f_n-\hat{f}_n$	$e_n=f_n-\hat{f}_n$	The error $e_n=f_n-\hat{f}_n$ is calculated and is fed to the <i>Symbol Encoder</i> , which
penev_B_F00386	11	1.000	e_n	e_n	error signal e_n . This error is added to the current reconstruction $\hat{f}_n$, which
penev_B_F00387	11	1.000	$\hat{f}_n=\hat{f}_n+e_n$	$\hat{f}_n=\hat{f}_n+e_n$	The result, $\hat{f}_n=\hat{f}_n+e_n$, is an identical copy of f_n .
penev_B_F00388	11	0.934	$f^{\text {rec }}\left(t_n\right)=\hat{f}_n$	$f^{\text {rec }}\left(t_n\right)=\hat{f}_n$	mates of the of the current value, $f^{\text {rec }}\left(t_n\right)=\hat{f}_n$. The encoder, which has access
penev_B_F00389	11	0.824	$e_n=f_n-\hat{f}_n$	$e_n=f_n-\hat{f}_n$	to the true value f_n , calculates the error $e_n=f_n-\hat{f}_n$ and feeds it, via the com-
penev_B_F00390	11	0.999	$t_n,\left\{f_n(m)=f\left(t_{n-m}\right)=f_{n-m}\right\}_{m=1}^M$	$t_n,\left\{f_n(m)=f\left(t_{n-m}\right)=f_{n-m}\right\}_{m=1}^M$	only a small time window, <i>preceding</i> t_n , $\left\{f_n(m)=f\left(t_{n-m}\right)=f_{n-m}\right\}_{m=1}^M$. The
penev_B_F00391	11	0.995	$\hat{f}_n=\alpha^m f_{n-m}$	$\hat{f}_n=\alpha^m f_{n-m}$	similar to (5.17)— $\hat{f}_n=\alpha^m f_{n-m}$.
penev_B_F00392	11	1.000	α_m	α_m	There are many possible choices for the coefficients α_m ; when they are
penev_B_F00393	11	0.998	$E\left\{e_n^2\right\}$	$E\left\{e_n^2\right\}$	chosen to minimize the expectation of the square of the error signal, $E\left\{e_n^2\right\}$, the
penev_B_F00394	11	1.000	$\left\{t_{n-m}\right\}_{m=1}^{M-1}$	$\left\{t_{n-m}\right\}_{m=1}^{M-1}$	is known, $\left\{t_{n-m}\right\}_{m=1}^{M-1}$, with $\left\{\mathbf{x}_m\right\}_{m=1}^{M-1}$, and the currently reconstructed grid
penev_B_F00395	11	1.000	$\left\{\mathbf{x}_m\right\}_{m=1}^{M-1}$	$\left\{\mathbf{x}_m\right\}_{m=1}^{M-1}$	is known, $\left\{t_{n-m}\right\}_{m=1}^{M-1}$, with $\left\{\mathbf{x}_m\right\}_{m=1}^{M-1}$, and the currently reconstructed grid
penev_B_F00396	11	0.964	$\mathbf{x}_M$	$\mathbf{x}_M$	point t_n , with $\mathbf{x}_M$. Then $\mathbf{Q}$ (5.40) is the correlation matrix of the signal,
penev_B_F00397	12	1.000	Φ	Φ	of experimentally well-controlled sensory stimuli Φ , interpreted as the horizon-
penev_B_F00398	12	1.000	$v(t)$	$v(t)$	tal velocity $v(t)$; simultaneously, the spike train $\left\{t_m\right\}$ that results from this
penev_B_F00399	12	1.000	$\left\{t_m\right\}$	$\left\{t_m\right\}$	tal velocity $v(t)$; simultaneously, the spike train $\left\{t_m\right\}$ that results from this
penev_B_F00400	12	0.702	$A^M(t)=F_1\left(t-t_M\right)$	$A^M(t)=F_1\left(t-t_M\right)$	enforced, $A^M(t)=F_1\left(t-t_M\right)$, and the m.s.e., $E\left\{\left(v(t)-v^{\text {rec }}(t)\right)^2\right\}$, is minimized
penev_B_F00401	12	0.702	$E\left\{\left(v(t)-v^{\text {rec }}(t)\right)^2\right\}$	$E\left\{\left(v(t)-v^{\text {rec }}(t)\right)^2\right\}$	enforced, $A^M(t)=F_1\left(t-t_M\right)$, and the m.s.e., $E\left\{\left(v(t)-v^{\text {rec }}(t)\right)^2\right\}$, is minimized
penev_B_F00402	12	1.000	$F(\tau)$	$F(\tau)$	over the experimental data set, the best $F(\tau)$ is determined. It is used to
penev_B_F00403	12	0.946	$v^{\text {rec }}(t)$	$v^{\text {rec }}(t)$	calculate the mutual information between $v(t)$ and $v^{\text {rec }}(t)$, which is a lower
penev_B_F00404	12	1.000	$A^m(t)$	$A^m(t)$	lar to (5.36), there are two major differences. The reconstructors $A^m(t)$ are
penev_B_F00405	12	0.424	t_M .	t_M .	identical—they are based on a single grid point, t_M .
penev_B_F00406	12	0.974	100%.	100%.	about the actual signal will be much higher, close to 100%.
penev_B_F00407	12	0.974	$^{36} \text {An}$	$^{36} \text {An}$	$^{36} \text {An}$ attempt to base them on two grid points, t_M and t_{M-1} , albeit also in a time-
penev_B_F00408	12	0.974	t_M	t_M	$^{36} \text {An}$ attempt to base them on two grid points, t_M and t_{M-1} , albeit also in a time-
penev_B_F00409	12	0.974	t_{M-1}	t_{M-1}	$^{36} \text {An}$ attempt to base them on two grid points, t_M and t_{M-1} , albeit also in a time-
penev_B_F00410	12	0.998	5–15%	5–15%	(5–15%, depending on the system under study).
penev_B_F00411	12	0.996	$\mathcal{P}_{\text {objects }}$	$\mathcal{P}_{\text {objects }}$	of the probability densities of images, $\mathcal{P}_{\text {images }}$, and of objects, $\mathcal{P}_{\text {objects }}$. We
penev_B_F00412	12	0.964	$\mathcal{P}_{\alpha}$	$\mathcal{P}_{\alpha}$	understood the probability distributions $\mathcal{P}_{\alpha}$ and $\mathcal{P}_{\text {objects }}$ as part of the explicit

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penev_B_F00413	12	■ 0.515	$\mathcal{P}_{\{\text{imges}\}}$	$\mathcal{P}_{\text{imges}}$	parameterization of $\mathcal{P}_{\text{images}}$.
penev_B_F00414	12	■ 0.800	$\mathcal{P}_{\{\text{objets}\}}$	$\mathcal{P}_{\text{objects}}$	the re-parameterization of $\mathcal{P}_{\text{objects}}$ with $\mathcal{P}_{\theta}$ and $\mathcal{P}_{\text{feature}}$. This hierarchical pa-
penev_B_F00415	13	■ 0.989	$O^{\{\text{rec}\}}(\mathbf{x}) = \mathcal{R}_{\mathcal{O}}^{M-1}(\mathbf{x})$	$O^{\text{rec}}(\mathbf{x}) = \mathcal{R}_{\mathcal{O}}^{M-1}(\mathbf{x})$	identical reconstructions $O^{\text{rec}}(\mathbf{x}) = \mathcal{R}_{\mathcal{O}}^{M-1}(\mathbf{x})$; the encoder calculates the recon-
penev_B_F00416	13	■ 0.991	$\mathcal{D}_{\{\mathcal{O}\}}(\mathbf{x})$	$\mathcal{D}_{\mathcal{O}}(\mathbf{x})$	$\mathcal{D}_{\mathcal{O}}(\mathbf{x})$ was largest. Here, we pose the question, Is it possible to find another
penev_B_F00417	13	■ 0.945	$\mathcal{D}_{\{0\}}(\mathbf{x}_{\{\text{spike}\}}) = \theta$	$\mathcal{D}_{\mathcal{O}}(\mathbf{x}_{\text{spike}}) = \theta$	time the current reconstruction error is equal to the threshold, $\mathcal{D}_{\mathcal{O}}(\mathbf{x}_{\text{spike}}) = \theta$.
penev_B_F00418	13	■ 0.999	$\mathcal{D}_{\{0\}}(\mathbf{x})^2$	$\mathcal{D}_{\mathcal{O}}(\mathbf{x})^2$	the individual grid points, $\mathcal{D}_{\mathcal{O}}(\mathbf{x})^2$. On one hand, when θ is relatively large,
penev_B_F00419	13	■ 0.938	$\mathcal{R}_{\mathcal{O}}^M(\mathbf{x})$	$\mathcal{R}_{\mathcal{O}}^M(\mathbf{x})$	tion, $\mathcal{R}_{\mathcal{O}}^M(\mathbf{x})$, but also, a current expectation of the variance of the error,
penev_B_F00420	13	■ 0.867	$D(\mathbf{x})^2 = E\{\mathcal{D}_{\mathcal{O}}(\mathbf{x})^2\}$	$D(\mathbf{x})^2 = E\{\mathcal{D}_{\mathcal{O}}(\mathbf{x})^2\}$	$D(\mathbf{x})^2 = E\{\mathcal{D}_{\mathcal{O}}(\mathbf{x})^2\}$ (see discussion in footnote 32 on page 77). Then, θ may
penev_B_F00421	13	■ 0.495	$\theta(\mathbf{x} - \mathbf{x}_{\text{vpik}})$	$\theta(\mathbf{x} - \mathbf{x}_{\text{vpik}})$	decays with the time since the last spike, $\theta(\mathbf{x} - \mathbf{x}_{\text{vpik}})$.
penev_B_F00422	13	■ 0.508	$\mathcal{D}_{\{0\}}(\mathbf{x})$	$\mathcal{D}_{\mathcal{O}}(\mathbf{x})$	the timing, which generates an error in the decoder's estimate of $\mathcal{D}_{\mathcal{O}}(\mathbf{x})$ and,
penev_B_F00423	13	■ 0.842	$O^{\{\text{rec}\}}(\mathbf{x})$	$O^{\text{rec}}(\mathbf{x})$	therefore, of $O^{\text{rec}}(\mathbf{x})$, which depends on the <i>derivative</i> of $O(\mathbf{x})$. ³¹ One way to
penev_B_F00424	13	■ 0.842	$O(\mathbf{x})$. ³¹	$O(\mathbf{x})$. ³¹	therefore, of $O^{\text{rec}}(\mathbf{x})$, which depends on the <i>derivative</i> of $O(\mathbf{x})$. ³¹ One way to
penev_B_F00425	13	■ 1.000	Δt	Δt	is determined with precision Δt , we may consider that a spike can be triggered
penev_B_F00426	13	■ 1.000	$D(\mathbf{x})$	$D(\mathbf{x})$	compares the target function, for example $D(\mathbf{x})$, with the threshold $\theta(\mathbf{x})$. Ob-
penev_B_F00427	13	■ 1.000	$\theta(\mathbf{x})$	$\theta(\mathbf{x})$	compares the target function, for example $D(\mathbf{x})$, with the threshold $\theta(\mathbf{x})$. Ob-
penev_B_F00428	13	■ 0.985	$\mathbf{x}_{\{\text{m}\}}$	$\mathbf{x}_{\text{m}}$	the difference is positive at the moment of the spike, $\mathbf{x}_{\text{m}}$, but was negative the
penev_B_F00429	13	■ 0.999	$\mathbf{x}'$	$\mathbf{x}'$	preceding moment, $\mathbf{x}'$.
penev_B_F00430	13	■ 0.991	$\mathbf{q}_{\mathbf{x}_{\text{m}}} \in S$	$\mathbf{q}_{\mathbf{x}_{\text{m}}} \in S$	The grid location $\mathbf{x}_{\text{m}}$ determines a vector $\mathbf{q}_{\mathbf{x}_{\text{m}}} \in S$ (4.38), which in turn de-
penev_B_F00431	13	■ 1.000	$U_{\{\text{m}\}} \subset S$	$U_{\text{m}} \subset S$	termines a one-dimensional subspace $U_{\text{m}} \subset S$ (vertical line), with a projector
penev_B_F00432	13	■ 0.997	$\mathbf{Q}_{\mathbf{x}_{\text{m}}}$	$\mathbf{Q}_{\mathbf{x}_{\text{m}}}$	$\mathbf{Q}_{\mathbf{x}_{\text{m}}}$ (4.39).
penev_B_F00433	13	■ 0.893	$\phi = \Phi^* s$	$\phi = \Phi^* s$	For the example $\phi = \Phi^* s$, in Section 4.4.2 we interpreted the choice of the
penev_B_F00434	13	■ 0.737	$O(\mathbf{x}_{\text{m}}) = O_{\text{m}} = \langle \mathbf{q}_{\mathbf{x}_{\text{m}}}, s \rangle$	$O(\mathbf{x}_{\text{m}}) = O_{\text{m}} = \langle \mathbf{q}_{\mathbf{x}_{\text{m}}}, s \rangle$	grid point $\mathbf{x}_{\text{m}}$ as the measurement $O(\mathbf{x}_{\text{m}}) = O_{\text{m}} = \langle \mathbf{q}_{\mathbf{x}_{\text{m}}}, s \rangle$ (4.37) and we
penev_B_F00435	13	■ 0.863	$\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$	$\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$	introduced $\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$ (4.42) (light vertical arrow, ending at A).
penev_B_F00436	13	■ 0.961	$\mathbf{q}' \in S$	$\mathbf{q}' \in S$	Analogously, any non-selected grid location $\mathbf{x}'$ determines the vector $\mathbf{q}' \in S$
penev_B_F00437	13	■ 0.659	$\mathbf{q}_{\text{x}}$	$\mathbf{q}_{\text{x}}$	(along the light slanted arrow). Together with $\mathbf{q}_{\mathbf{x}_{\text{m}}}$, they determine a 2-
penev_B_F00438	13	■ 0.659	2-	2-	(along the light slanted arrow). Together with $\mathbf{q}_{\mathbf{x}_{\text{m}}}$, they determine a 2-
penev_B_F00439	13	■ 1.000	$U_{\{\text{m}\}}$	U_{m}	U_{m} and U'_{m} .
penev_B_F00440	13	■ 1.000	$U_{\{\text{m}\}}^{\{\text{prime}\}}$	U'_{m}	U_{m} and U'_{m} .
penev_B_F00441	13	■ 0.843	$N - 2$	$N - 2$	All other $N - 2$ dimensions of the PCA S -subspace are collapsed to $U^{\perp}$.
penev_B_F00442	13	■ 0.843	$U^{\{\text{perp}\}}$	$U^{\perp}$	All other $N - 2$ dimensions of the PCA S -subspace are collapsed to $U^{\perp}$.
penev_B_F00443	13	■ 0.958	$O_{\{\text{m}\}} \mathbf{p}'$	$O_{\text{m}} \mathbf{p}'$	For a given measurement O_{m} , we can define $O_{\text{m}} \mathbf{p}'$ (light slanted arrow, ending
penev_B_F00444	13	■ 1.000	$N - 1$	$N - 1$	at B). It can be treated as a normal to an $N - 1$ dimensional hyper-plane, whose
penev_B_F00445	13	■ 0.990	$(B - C)$	$(B - C)$	intersection with the plane of the circle is the slanted line $(B - C)$, tangential
penev_B_F00446	13	■ 1.000	C	C	to the circle, and which intersects with U'_{m} at point C . This hyper-plane's
penev_B_F00447	13	■ 0.914	$\phi = \Phi^* s$	$\phi = \Phi^* s$	the context of sparsification. Given the example $\phi = \Phi^* s$, and the choice
penev_B_F00448	13	■ 0.380	$\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$ (4.42)	$\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$ (4.42)	$\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$ (4.42), point A on Fig. 5.10, and a vector $\mathbf{s}_{\perp}$, orthogonal
penev_B_F00449	13	■ 0.380	$\mathbf{s}_{\perp}$	$\mathbf{s}_{\perp}$	$\mathbf{s}_{\parallel} = O_{\text{m}} \mathbf{p}_{\mathbf{x}_{\text{m}}} \in U_{\text{m}}$ (4.42), point A on Fig. 5.10, and a vector $\mathbf{s}_{\perp}$, orthogonal
penev_B_F00450	13	■ 0.829	$\mathbf{s}_{\parallel}$	$\mathbf{s}_{\parallel}$	to it. We interpreted $\mathbf{s}_{\parallel}$ as the maximum likelihood estimate.
penev_B_F00451	13	■ 1.000	$O(\mathbf{x}_{\text{m}}) = O_{\text{m}}$	$O(\mathbf{x}_{\text{m}}) = O_{\text{m}}$	decoder cannot assume that $O(\mathbf{x}_{\text{m}}) = O_{\text{m}}$ exactly, but only that $O(\mathbf{x}_{\text{m}}) \geq O_{\text{m}}$.
penev_B_F00452	13	■ 1.000	$O(\mathbf{x}_{\text{m}}) \geq O_{\text{m}}$	$O(\mathbf{x}_{\text{m}}) \geq O_{\text{m}}$	decoder cannot assume that $O(\mathbf{x}_{\text{m}}) = O_{\text{m}}$ exactly, but only that $O(\mathbf{x}_{\text{m}}) \geq O_{\text{m}}$.
penev_B_F00453	13	■ 0.989	$O(\mathbf{x}')$	$O(\mathbf{x}')$	positive transition—that the value $O(\mathbf{x}')$, immediately prior to the spike, was
penev_B_F00454	13	■ 0.822	O' . ³³	O' . ³³	below a certain threshold, O' . ³³ We can decompose $\mathbf{s}$ into a sum of three mutu-
penev_B_F00455	13	■ 1.000	$\mathbf{s}_{\parallel} \in U_{\text{m}}, \mathbf{s}' \in U'_{\text{m}}$	$\mathbf{s}_{\parallel} \in U_{\text{m}}, \mathbf{s}' \in U'_{\text{m}}$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_{\text{m}}, \mathbf{s}' \in U'_{\text{m}}$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_{\text{m}}}$
penev_B_F00456	13	■ 1.000	$\mathbf{s}_{\perp} \in U^{\perp}$	$\mathbf{s}_{\perp} \in U^{\perp}$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_{\text{m}}, \mathbf{s}' \in U'_{\text{m}}$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_{\text{m}}}$
penev_B_F00457	13	■ 1.000	$\mathbf{q}'$	$\mathbf{q}'$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_{\text{m}}, \mathbf{s}' \in U'_{\text{m}}$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_{\text{m}}}$
penev_B_F00458	13	■ 1.000	$\mathbf{q}_{\mathbf{x}_{\text{m}}}$	$\mathbf{q}_{\mathbf{x}_{\text{m}}}$	ally orthogonal vectors, $\mathbf{s}_{\parallel} \in U_{\text{m}}, \mathbf{s}' \in U'_{\text{m}}$, and $\mathbf{s}_{\perp} \in U^{\perp}$. Since both $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_{\text{m}}}$
penev_B_F00459	13	■ 0.999	$U^{\{\text{perp}\}}, \mathbf{s}_{\perp}$	$U^{\perp}, \mathbf{s}_{\perp}$	are orthogonal to $U^{\perp}$, $\mathbf{s}_{\perp}$ can be anywhere in $U^{\perp}$, and a maximum likelihood
penev_B_F00460	13	■ 0.972	$\mathbf{s}_{\perp} = \mathbf{0}$	$\mathbf{s}_{\perp} = \mathbf{0}$	estimate of it is $\mathbf{s}_{\perp} = \mathbf{0}$.
penev_B_F00461	13	■ 0.989	$\mathbf{s}'$	$\mathbf{s}'$	On the other hand, $\mathbf{s}'$ can be anywhere on the ray in U'_{m} , starting at C ,
penev_B_F00462	13	■ 0.998	$\mathbf{s}' = \mathbf{0}$	$\mathbf{s}' = \mathbf{0}$	which is the mean of the respective distribution, is not $\mathbf{s}' = \mathbf{0}$, but is to the
penev_B_F00463	13	■ 0.994	D	D	left of A , at D . Also, the line $B - C$ intersects with U_{m} at some place, and that
penev_B_F00464	13	■ 0.994	BC	BC	left of A , at D . Also, the line $B - C$ intersects with U_{m} at some place, and that

Identifier	Page	Conf.	LaTeX source	Rendered	Image
penev_B_F00465	13	■ 0.943	$\mathbf{o}_m = \left\langle \left\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \right\rangle, \mathbf{s}_{\parallel} \right\rangle$	$o_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$	is an upper bound for $\underline{b}_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$; hence the maximum likelihood estimate
penev_B_F00466	13	■ 0.952	$0_m \setminus \mathrm{p}_m(A)$	$O_m \mathbf{p}_m(A)$	for $\mathbf{s}_{\parallel}$ is not $O_m \mathbf{p}_m(A)$, but is a bit higher. ³⁴
penev_B_F00467	13	■ 0.350	$\mathbf{a}_m = \left\langle \left\langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \right\rangle, \mathbf{s}_{\parallel} \right\rangle$	$a_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$	³² The values of the projection $\underline{b}_m = \langle \mathbf{q}_{\mathbf{x}_m}, \mathbf{s}_{\parallel} \rangle$ will be distributed according to the orig-
penev_B_F00468	13	■ 0.966	$o_m \geq 0_m$	$o_m \geq O_m$	inal probability distribution, uormal by assumption (2.49), <i>conditional</i> on $o_m \geq O_m$. The
penev_B_F00469	13	■ 0.553	$0_m \setminus \mathrm{p}_m$	$O_m \mathbf{p}_m$	³³ For simplicity, $O_m \mathbf{p}_m$ and $O' \mathbf{p}'$ are shown to have the same lengths on Fig. 5.10. In
penev_B_F00470	13	■ 0.553	$0^{\prime \prime} \setminus \mathrm{p}^{\prime \prime}$	$O' \mathbf{p}'$	³³ For simplicity, $O_m \mathbf{p}_m$ and $O' \mathbf{p}'$ are shown to have the same lengths on Fig. 5.10. In
penev_B_F00471	13	■ 0.810	$\mathbf{q}^{\prime \prime}$	$\mathbf{q}'$	probability distribution along U'_m ; the smaller the angle α between $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$, the more
penev_B_F00472	13	■ 0.810	$\mathbf{q}_{\mathbf{x}_m}$	$\mathbf{q}_{\mathbf{x}_m}$	probability distribution along U'_m ; the smaller the angle α between $\mathbf{q}'$ and $\mathbf{q}_{\mathbf{x}_m}$, the more
penev_B_F00473	13	■ 1.000	$C \rightarrow A$	$C \rightarrow A$	informative the lack of a spike. Ultimately, when $C \rightarrow A$, half of the axis U'_m is excluded,
penev_B_F00474	13	■ 0.848	o_m	o_m	o_m approaches O_m , and the information due to the localization of $\mathbf{s}_{\parallel}$ in U_m increases.
penev_B_F00475	13	■ 0.848	$s_{\parallel}$	$s_{\parallel}$	o_m approaches O_m , and the information due to the localization of $\mathbf{s}_{\parallel}$ in U_m increases.
penev_B_F00476	13	■ 0.810	$\cos \alpha = P_m(x')$	$\cos \alpha = P_m(x')$	This is intuitively correct; $\cos \alpha = P_m(\mathbf{x}')$, the output correlation between two adjacent
penev_B_F00477	14	■ 0.867	$\mathcal{P}_{\text{world}}$	$\mathcal{P}_{\text{world}}$	parameterization of the probability density $\mathcal{P}_{\text{world}}$. Using this parameterization,
penev_B_F00478	14	■ 1.000	$T=1038$	$T = 1038$	We selected from the FERET database $T = 1038$ examples without glasses
penev_B_F00479	14	■ 1.000	64×60	64×60	that line. Finally, the examples in the data set were cropped through a 64×60
penev_B_F00480	14	■ 0.973	$r(\theta, \ell)$	$r(\theta, \ell)$	ordinates $r(\theta, \ell)$ with 512 samples spanning a full revolution in the θ direction
penev_B_F00481	14	■ 1.000	ℓ	ℓ	and 256 samples in the ℓ direction. Anthropological landmarks on the surfaces
penev_B_F00482	14	■ 0.823	$\theta_{\text{fixed}}, \ell_{\text{fixed}}$	$\theta_{\text{fixed}}, \ell_{\text{fixed}}$	The examples were aligned by us, so that the fixed point $(\theta_{\text{fixed}}, \ell_{\text{fixed}})$
penev_B_F00483	14	■ 0.998	3×3	3×3	smoothed with a 3×3 Gaussian filter and under-sampled twice to produce a
penev_B_F00484	14	■ 0.981	256×128	256×128	256×128 representation. All examples in the <i>Mini Survey study</i> database were
penev_B_F00485	14	■ 1.000	$T=348$	$T = 348$	chosen to produce a $T=348$ data set.
penev_B_F00486	14	■ 0.999	128×64	128×64	Finally, the examples in the data set were cropped through a 128×64
penev_B_F00487	14	■ 1.000	96×90	96×90	cropped through a 96×90 window centered horizontally about the fixed point
penev_B_F00488	14	■ 1.000	96×120	96×120	cropped through a 96×120 window centered horizontally about the fixed point
penev_B_F00489	14	■ 1.000	$\left\{ s_n = 1.05^{n/2} \right\}_{n=-4}^{+4}$	$\left\{ s_n = 1.05^{n/2} \right\}_{n=-4}^{+4}$	ing parameter $\{s_n = 1.05^{n/2}\}_{n=-4}^{+4}$, i.e., the up and down deviations from the
penev_B_F00490	14	■ 1.000	$\approx \{10\%, 7.5\%, 5\%, 2.5\%, 0\%\}$	$\approx \{10\%, 7.5\%, 5\%, 2.5\%, 0\%\}$	hand-determined scale were $\approx \{10\%, 7.5\%, 5\%, 2.5\%, 0\%\}$, resulting in dimen-
penev_B_F00491	14	■ 0.982	$T=10944, V=3840$	$T = 10944, V = 3840$	sionalities ($T = 10944, V = 3840$).
penev_B_F00492	14	■ 0.983	$\left\{ s_m = 1.025^{m/3} \right\}_{m \in \{-2, -1, 1, 2\}}$	$\left\{ s_m = 1.025^{m/3} \right\}_{m \in \{-2, -1, 1, 2\}}$	additional scalings $\{s_m = 1.025^{m/3}\}_{m \in \{-2, -1, 1, 2\}}$, resulting in deviations of $\approx$
penev_B_F00493	14	■ 0.983	$\approx$	$\approx$	additional scalings $\{s_m = 1.025^{m/3}\}_{m \in \{-2, -1, 1, 2\}}$, resulting in deviations of $\approx$
penev_B_F00494	14	■ 0.651	$V=96 \times 120=11,520$	$V = 96 \times 120 = 11,520$	Three croppings were prepared—full with $\underline{V = 96 \times 120 = 11,520}$, b3q
penev_B_F00495	14	■ 0.992	$V=96 \times 90=8,640$	$V = 96 \times 90 = 8,640$	with $\underline{V = 96 \times 90 = 8,640}$ which is the same as that for Ensemble 3, and with
penev_B_F00496	14	■ 0.881	$V=64 \times 60=3,840$	$V = 64 \times 60 = 3,840$	third $\underline{V = 64 \times 60 = 3,840}$ is the same as Ensemble 1.
penev_B_F00497	14	■ 0.992	$10 \times$	$10 \times$	The image was converted to grayscale, oversampled $10 \times$, and then under-